

Cryptocurrencies and Interest Rates: Inferring Yield Curves in a Bondless Market

Philippe BERGAULT* Sébastien BIEBER† Olivier GUÉANT‡ Wenkai ZHANG§

Abstract

In traditional financial markets, yield curves are widely available for countries (and, by extension, currencies), financial institutions, and large corporates. These curves are used to calibrate stochastic interest rate models, discount future cash flows, and price financial products. Yield curves, however, can be readily computed only because of the current size and structure of bond markets. In cryptocurrency markets, where fixed-rate lending and bonds are almost nonexistent as of early 2025, the yield curve associated with each currency must be estimated by other means. In this paper, we show how mathematical tools can be used to construct yield curves for cryptocurrencies by leveraging data from the highly developed markets for cryptocurrency derivatives.

Keywords: Interest Rates, Yield Curves, Cryptocurrencies, Derivatives.

1 Introduction

1.1 Yield Curves: A Familiar Tool

Yield curves have become a fundamental component of market data on trading floors worldwide. Beyond very short-term interest rates set by central banks and the interbank market, yield curves reflect the borrowing conditions of bond issuers such as governments, major corporations, municipalities, etc. They play a crucial role in pricing simple fixed-income instruments across various currencies and serve as key inputs for calibrating stochastic interest rate models used to discount future cash flows and then price complex financial products.

The process of constructing a yield curve for a given issuer typically requires access to market prices for that issuer's bonds across a range of maturities. From these prices and the associated bond characteristics, a curve can be derived using bootstrapping techniques, followed by appropriate smoothing – often employing spline methods – or through penalized nonlinear regression approaches (see, for more details, the standard reference [BM01]).

For many countries, debt structures are currently such that estimating a yield curve is relatively straightforward. However, this has not always been the case, and it is important to be aware that, beyond mathematics, the ability of financial practitioners to discount future cash flows at precise interest rates is primarily a consequence of the debt emissions of countries and the evolution of bond markets in recent decades.

This observation raises a critical question: how should one proceed to construct a yield curve when a new currency emerges without the simultaneous development of a bond market? This issue is particularly relevant in the context of cryptocurrencies. Indeed, while markets for trading fixed-term futures, perpetual futures, and options have developed on major cryptocurrencies, a bond market for fixed-rate lending in these new currencies across different maturities remains largely nonexistent as of early 2025.

*Université Paris Dauphine-PSL, Ceremade, Paris, France, bergault@ceremade.dauphine.fr.

†Université Paris Dauphine-PSL, Ceremade, Paris, France, bieber@ceremade.dauphine.fr.

‡Université Paris 1 Panthéon-Sorbonne, Centre d'Economie de la Sorbonne, Paris, France, olivier.gueant@univ-paris1.fr.

§LCH SA, Paris, France, wenkai.zhang@lseg.com.

The goal of this paper is to propose an alternative to classical yield curve construction methods, based on derivatives prices, that could be applied to cryptocurrency markets. Before presenting our approach, however, we provide a historical perspective on interest rates and yield curves.

1.2 Historical Perspectives

The concept of the yield curve is, in fact, a modern one, and yield curves could hardly be computed before the second half of the 20th century, except for some very specific countries. This is, of course, not due to any novelty in the concept of credit or interest rates. Indeed, whether formal or informal, with or without the payment of interest, debts and credits have been integral to human civilizations since time immemorial – long before the invention of money and, according to [Gra14], of greater historical importance than barter.

1.2.1 Ancient times

In the ancient Near East, old Sumerian documents from circa 3000 BCE reveal a widespread use of credit with interest, including loans of grain measured by volume and loans of metal (silver) measured by weight. At the Louvre Museum in Paris, the Cone of Entemena, dating to around 2400 BCE, mentions interest-bearing loans. Also housed at the Louvre, the famous Code of Hammurabi, circa 1750 BCE – a Babylonian compilation and revision of earlier Sumerian and Akkadian law codes – explicitly specifies interest rate figures:

“If a merchant lends grain at interest, for one *gur* he shall receive one hundred *sila* as interest; if he lends money at interest, for one shekel of silver he shall receive one-fifth of a shekel as interest.”

This corresponds to interest rates of $33\frac{1}{3}\%$ for grain and 20% for silver.¹ These rates, which remained in place for more than 1,200 years, should be regarded as legal maxima applicable to both short-term loans (e.g., repayment after the harvest) and longer-term loans (with documented cases of three-year loans), as well as to both production and consumption loans. In practice, actual interest rates could be lower, yet they still tended to be in the two-digit range (see [SH13]).²

From the Babylonians, we can move to the Greeks who were the first to mint coins in the 7th century BCE and developed a new economic and commercial system. From ship loans to pawn credits, credit and debt played a significant role in ancient Greek society, as evidenced indirectly by Solon’s radical reforms in the 6th century BCE: through his *seisachtheia*, Solon reduced debts, abolished debt slavery,³ and emancipated those who had fallen into servitude for this reason.

Regarding the characteristics of loans in classic Greece, loans to finance maritime trade were very common (see [Coh97]) and could yield interest rates between 20% and 30% per voyage (not annual basis). Personal loans could be secured by movable assets (pawnshops) or by land (*hypothèque*) and, according to [SH13], typically had relatively short durations, say up to a year or so, and involved monthly interest payments.

Loans to states or cities were relatively rare before the 3rd or 2nd century BCE and were secured by wealthy citizens (underwriters), *hypothèques*, or public revenues.⁴ For example, in the 4th century BCE, Demosthenes lent to the city of Oreus at 12%, with the credit secured by the city’s public revenues (see [SH13]).

Sources mention interest rates in the 16%–18% interval at the time of Solon, followed by a gradual decline from 10%–12% in the 5th century BCE to 6%–10% by the end of the 2nd century BCE. Unlike in Babylonian times, we often know the purpose of the loans and the collateral. Yet, it remains impossible to construct a classification of interest rates as a function of loan’s characteristics: purpose, term, debtor’s reputation, etc.

¹These figures were probably chosen for their simplicity within the local system of fractional arithmetic (see [HdM02]).

²It is known that temples played a major role in credit markets. First, legal documents regarding loans were often deposited in temples. Second, temples themselves loaned metals and grains, often charging interest rates below the legal maximum.

³Debt slavery and debt cancellations are frequently mentioned in ancient scriptures; see Leviticus 25, for instance.

⁴Sometimes, also, state loans were compulsory.

In Rome, from the 5th century BCE, with the promulgation of the Twelve Tables in 449 BCE, until the first century, the legal limit on interest rates was set – except for a few years – at $8\frac{1}{3}\%$, and it is thought that most loans at the time were short-term and often secured by real estate (see [SH13]).⁵

The data becomes far richer from the first century BCE (see [And01]), which begins with an increase in the legal maximum, set at 12% in 88 BCE. Nevertheless, prevailing interest rates were often well below this ceiling, especially during times of peace. As noted in [Tem04], loans were sufficiently widespread at the time to allow for general discussions about interest rates and standard lending conditions. In his *Letters to Atticus*, Cicero commented on an increase in interest rates from $\frac{1}{3}\%$ to $\frac{1}{2}\%$ per month. In *The History of Rome*, Livy wrote:

“As long as the succeeding consuls – T. Manlius Torquatus and C. Plautius – held office, the same peaceful conditions prevailed. The rate of interest was reduced by one half, and payment of the principal was to be made in four equal installments [...].”

In the first century, normal interest rates in Rome typically ranged from 4% to 6%, except in times of crisis, when they could rise to the legal maximum (and occasionally, though rarely, exceed it). As in the previous century, loans were common: for instance, Columella’s *De re rustica* (on agriculture) offers advice to those setting up vineyards, including how to manage costs related to interest payments. Loans were often made between private individuals, or provided by government-supervised groups (*argentarii*, *nummularii*, etc.), and there were no large commercial banks – in particular, no prominent banker name has survived to the present day. However, the existence of financial intermediaries, often linked to religious institutions, is clearly documented. These intermediaries accepted deposits (mainly term deposits) and provided credit to borrowers by pooling financial resources.

After the first century, the legal limit remained at 12% until an increase of $\frac{1}{2}\%$ in the 4th century, suggesting that the former legal limit was often reached in actual loans at that time. In reality, the relatively low interest rates of the first century would not be seen again in Western Europe for more than one thousand years.

Whether for the Babylonians, Greeks, or Romans, it appears that there is no possibility of constructing yield curves. Although we have occasional traces of loans lasting two to five years, most loans were for a year or less. Moreover, there was no (secondary) market enabling lenders to transfer loan positions before maturity, and thus no mechanism to reveal on a regular basis the dynamics of credit supply and demand.

1.2.2 Medieval Period

The evolution of credit and interest rates in medieval Europe is inseparable from the Christian doctrine on usury. In the Old Testament, we read in Deuteronomy 23:19–20:

“You shall not charge interest on loans to your brother, interest on money, interest on food, interest on anything that is lent. You may charge a stranger interest, but you may not charge your brother interest ...”

However, with the notion of universal brotherhood introduced by Christianity, there was no longer any “stranger,” and lending at interest naturally came to be regarded as morally unacceptable in the Christian world. The Church began prohibiting usury for clerics as early as the Council of Nicaea, and later discouraged laypeople from engaging in interest-bearing lending, eventually escalating sanctions by resorting to excommunication (though pawnbrokers were often not Christian). As for state and civil law, the Capitularies of Charlemagne forbade lending at interest for everyone. Even if the enforcement of such laws was not always consistent, and pawnshops were often considered a necessary evil, most people in the Christian medieval world feared being found guilty of usury.

In parallel with the influence of Christianity, economic activity in Western Europe remained extremely limited during the early Middle Ages. In the 8th and 9th centuries, due to restricted access to the sea – Arab pirates in the south and Viking incursions in the north – maritime trade was severely constrained,

⁵Maritime loans were not subject to this legal maximum.

and commerce remained mostly local, in a society increasingly centered on agriculture and structured by feudalism. By the end of the 10th century and into the eleventh, commerce gradually began to revive in Europe. Venice, which had never fully ceased commercial activity, saw its role expand through agreements with Constantinople and exchanges with the Muslim world. Other cities, especially in northern Italy and across northern Europe, also gained commercial importance as they secured greater autonomy, resulting in the emergence of a merchant class.

It was really in the 12th century that a form of commercial economy began to develop, and wealthy merchants started to act as bankers by issuing commercial loans with maturities often aligned with major trade fairs (such as the important Champagne fairs). These loans typically lasted a few months and, for the religious reasons outlined previously, interest payments were often concealed in currency conversions using bills of exchange. For the late 12th century, [SH13] documents interest rates of around 20% in Genoa and between 10% and 16% in what is now the Netherlands.

The 12th century also saw the emergence of wealthy families who could invest in new debt instruments taking the form of life or perpetual annuities, secured by property. The interest payments on these instruments were not considered usury because the principal was not necessarily repaid. Rates in the range of 8% to 10% are reported in [SH13] for such long-term loans. Forced loans imposed by cities, such as the *prestiti* in Venice, constituted another form of long-term debt that would play an important role in the centuries to come.

Interestingly, this period marks the beginning of a term structure of interest rates: short-term maturities reflecting the calendar of trade fairs and a point at infinity on the yield curve with the creation of annuities and perpetual debts – forms of credit shaped by religious constraints. However, the debtors at the two ends of the spectrum were not the same.

The 13th century was first and foremost a century of Scholastic thought, and in particular, the century of St. Thomas Aquinas. It was during this period that exceptions to the canonical prohibition of usury were formally theorized, in order to distinguish usury from licit interest. One of the central ideas was that interest could be permitted insofar as it did not constitute a profit, but rather compensation for a loss: for example, interest due to delayed repayment could be considered legitimate, as could the payment of a fee to cover the time and effort involved in issuing and managing the loan, and so on.

This century also saw the flourishing of international trade, with important economic centers such as Genoa, Bruges, Cologne, and even London (for wool trade). The expansion of commerce was supported by new financial innovations led by the Italians, especially the Florentines,⁶ who began to operate as bankers on a continental scale. [SH13] reports double-digit interest rates on certain deposits and on most loans during this period, with a general downward trend over the course of the century. It also witnessed the expansion of annuities and perpetual loans, which in some cases became tradable instruments. A notable example is the Venetian *prestiti* – although associated with forced loans – initially issued with a nominal rate of 5%.

The 13th century somehow offers a glimpse of what finance would come to resemble in the early modern period. However, the 14th century marked a pause, due to a slowdown in economic activity and, more significantly, the Black Death – which decimated between 30% and 60% of the European population in the middle of the century – and the Hundred Years' War, which would continue until the mid-15th century.

In spite of these events, data on loans and interest rates during the 14th and early 15th centuries is more abundant than in previous periods. It notably includes a price chronicle of the Venetian *prestiti*, which allows us, for the first time in history and over an extended period,⁷ to trace the evolution of long-term interest rates – in a sense, the long-term limit of the still non-existent yield curve – associated with the debt of a single issuer.

⁶Not to be confused with the Lombards, who had long been active in the pawnshop business and charging very high rates.

⁷For the early 15th century, converting prices into interest rates is more difficult, since interest payments were often suspended for conjunctural reasons.

1.2.3 Early Modern Europe

The period following the end of the Hundred Years' War was one of discoveries and inventions, during which commerce once again expanded across Europe and would soon extend beyond. The 15th century was also that of the Medici Bank and of several other banking institutions offering interest on deposits (see [SH13] for figures). During the second half of the century, public pawnshops were established to replace part of the lending activity previously dominated by loan sharks, who had charged very high interest rates to the poor for centuries. Interestingly, the low but nonzero rates charged by these public institutions were not considered usurious, as they were intended to cover operational costs.

Taking over the financial leadership from Italy, Antwerp (then under Spanish control) became, until its default in 1570, the major financial center of Europe. Its Bourse served as a hub where bankers, financial agents, and merchants exchanged all types of credit instruments, including bills of exchange used in international trade.

Before the 16th century, princes were often heavily indebted and recurrently rolled over short-term loans at rates higher than those for commercial credit, due to their poor creditworthiness. In the case of the Spanish Netherlands government, [SH13] documents more than twenty short-term loans (most with maturities well below one year) negotiated between 1509 and 1521, often in Antwerp or Bruges. Interest rates varied considerably during this period: for example, an 18-month loan at $7\frac{1}{2}\%$ in June 1510, a 4-month loan at 24% in the same month, and a 10-month loan at $6\frac{1}{4}\%$ less than a year later. Clearly, despite the relatively high number of observations, there is no way to construct the first segment of a yield curve in any consistent manner.

By the end of the 16th century, the model of long-term borrowing through life or perpetual annuities had been widely adopted by cities such as Genoa,⁸ Barcelona, Nuremberg, and Amsterdam, as well as by states, including Francis I's France, which began issuing *rentes*, and the Spanish Crown, which refinanced its floating debt through perpetuums. Fixed-maturity long-term loans remained very rare, but one notable example is that of the French king Henri II, who refinanced the short-term rolling debt inherited from the wars of his father (with Charles V, Holy Roman Emperor) by raising new funds through an 11-year loan in 1555, known as the *Grand Parti de Lyon*.

The 16th century also marked the beginning of a shift in perspective regarding usury, following the Protestant Reformation. While Luther remained aligned with Scholastic thinking on the subject, this was not the case for Calvin, who argued that charging interest on commercial loans could be acceptable. Beyond theological debates, England under Henry VIII broke with the longstanding prohibition and introduced a legal maximum interest rate, which would be regularly updated over time and for which more and more exceptions will be later discussed (see [Mun12]).

The 17th century was a century of contrasts.⁹ It was both a century of expanding trade – marked by the rise of major commercial companies and the development of several exchanges across Europe, including the Amsterdam Bourse, which became a major financial center – and a century of frequent wars that brought severe financial stress, including several French defaults and repeated bankruptcies by the Spanish Crown, despite the wealth expected to flow from Latin America.

In terms of credit structure, governments financed themselves both through short-term borrowing – often at high interest rates – and through very long-term instruments, as debt was increasingly restructured in the form of perpetual loans (such as the *rentes* in France), life annuities (e.g., in the Dutch Republic and, at the very end of the century, in England), and some rare long-term loans (exceeding 30 years in the case of Holland).

⁸The *luoghi* in Genoa are slightly different because payments were not fixed and should be regarded as dividends.

⁹We focus here on state borrowing, but commercial loans were of course very common. One cannot speak of lending in early modern Europe without thinking of Shylock, the character from Shakespeare's *The Merchant of Venice*, who lends money at interest – or, in the story, against “a pound of flesh.” Merchants typically provided short-term credit directly to their suppliers and clients. Long-term private loans were nearly nonexistent, except in the form of land mortgages.

By the 17th century, interest rates were becoming increasingly national in character, with particularly low rates (as low as 3%) and minimal collateral requirements in Holland. Despite the active presence of governments on both the short-term and very long-term segments, it remains difficult to compute a clear yield curve for at least two reasons: (i) short-term rates were highly volatile and appeared to be only weakly linked to default risk or the quality of collateral; and (ii) although secondary markets existed for long-term debt, this was not yet the case for non-commercial short-term instruments, making it difficult to form a consistent temporal view of interest rate dynamics beyond long-term ones.

1.2.4 The 18th Century

If the Dutch had taken over from the Italians in the preceding centuries, the 18th century marks the beginning of London’s financial supremacy (see [Ash13]). The Amsterdam market remained active, but it was periodically marked by episodes of overspeculation, while London gradually took the lead with its more stable money market. The Bank of England, founded at the very end of the 17th century, played a central role by setting discount rates for both inland and foreign bills, with typical rates ranging between 3% and 6% throughout the 18th century and exhibiting rather low volatility.¹⁰ Decades later, it inspired the creation of the *Caisse d’Escompte* under Louis XVI in France, which issued notes and extended short-term loans to the government at 4%.

With regard to government borrowing, England borrowed recurrently during the 18th century, mainly through perpetual loans and various types of annuities – often incorporating lottery features (see [Coh53]). Benefiting from declining interest rates, England refinanced its debt by issuing consolidated annuities in the middle of the century, and soon introduced the 3% consols: perpetual loans with a 3% coupon, redeemable at the issuer’s discretion. Two major British innovations – standardization and transparency in public finance – contributed to the development of an active secondary market. The prices of 3% consols can be followed until 1888, when they were exchanged for lower-coupon consols. For England, the long-term limit of the yield curve has thus been observable since the mid-18th century.

The French and Dutch cases were markedly different, characterized by a lack of standardization and transparency. French *rentes* came in a wide variety of forms – perpetuums, life annuities, and even up to four-life annuities – although many (issued at the beginning of the Seven Years’ War) carried a 5% coupon.¹¹ Dutch obligations also took various forms: there were multiple issuers beyond the Dutch Republic itself, including towns and state-backed organizations, and a wide range of terms. [SH13] documents perpetuums, life annuities, and fixed-term bonds of 30 and 32 years. Fragmentation and lack of transparency prevented the emergence of an active secondary market.

At the end of the 18th century, the United States gained independence from England and, as early as 1776 and 1777, Congress authorized the issuance of medium-term domestic debt (3-year maturity) at interest rates of 4% and 6%. The American Revolution and the early years of the United States were also financed through loans obtained in Europe – most notably from France and the Netherlands – with maturities ranging from 10 to 25 years. Notably, no perpetual debt was issued during this early phase.

A foundational moment in the history of American public finance occurred with Hamilton’s restructuring of federal and state debts at the end of the century. As Secretary of the Treasury, he consolidated existing obligations and issued new perpetual bonds – redeemable at the discretion of the government – thereby establishing the credibility of the federal government.

¹⁰From the middle of the 18th century, the financial system does not only finance commerce but also starts financing industrial activities.

¹¹Beyond financial market development, the disorder in French public finance was so significant that the accumulation of debt over the 18th century became one of the major contributing factors to the outbreak of the French Revolution in 1789.

1.2.5 The 19th Century

Europe

Throughout the 19th century, the vast majority of British national debt consisted of consols, with a sustained effort toward standardization through the conversion of higher-coupon consols into 3% consols. In 1888, owing to persistently low interest rates and the government's right to redeem the debt at its discretion, the 3% consols were converted into consols with even lower coupons.

The 19th century also marked the beginning of the modern era in British public debt issuance, with the introduction of fixed-maturity bonds. The first such instruments – exchequer bonds – were issued in 1853 with a 40-year maturity and low coupons, but they met with limited success. More significant and popular were the multiple exchequer bond issuances of the late 1870s, this time with 3-year maturities. On the short end of the yield curve, exchequer bills began to be issued from 1877 onwards, typically with 3-month maturities, and occasionally with 6- or 12-month terms.

By the end of the century, a British yield curve – albeit with only a few key points – could thus be reconstructed. It is particularly noteworthy that short-term interest rates exhibited high volatility throughout the century, whereas long-term rates were far more stable and exhibited a structural decline.

In France, *rentes* remained extremely popular, although efforts toward standardization were less pronounced than in England. At various times, *rentes* with 5%, 4 $\frac{1}{2}$ %, 4%, and 3% coupons were all actively traded on the secondary market. Due to differing probabilities of redemption, their yields often diverged, though they tended to follow the same general trend as British consols – albeit with a positive credit spread.¹² Interestingly, while long-term rates were systematically higher in France than in England, the reverse held for short-term rates.

Dutch perpetuums and their yields exhibited similar patterns than their French counterparts: they followed the trajectory of British consols, again with a positive credit spread, but with greater volatility. As in France, Dutch short-term interest rates were generally lower than their British counterparts.

At the dawn of the 20th century, government borrowing in Europe was still primarily conducted through very long-term instruments – typically life annuities or perpetuums – with the principal potentially repayable only at the borrower's discretion. The structure of public debt beyond the very short term was thus clearly the product of a historical trajectory shaped by two intertwined forces: on the one hand, the Church's original doctrine on interest-bearing loans; and on the other, the legacy of a feudal system in which long-lasting income streams were granted to subjects as compensation for services.

What may seem most surprising is the enduring appetite of investors for such instruments given that perpetual bonds are, by design, highly sensitive to interest rate fluctuations. That said, questioning investors' preference for these products might be anachronistic, given the lack of alternatives at the time. Indeed, the figure of the *rentier* is a recurring one in the novels of Balzac and Zola, where characters are described by the income generated by their annuities rather than by the market value of their holdings.

Outside Europe

Unlike for European countries, many loan contracts from debtors outside Europe and in the British colonies – often traded in London or Paris and denominated in sterling or franc – included a maximum maturity date by which the principal had to be repaid or incorporated amortization features for gradual repayment. For a detailed list of issues in London in the 19th century, see [SH13].

The case of the United States is special and deserves interest as the global financial center would shift from London to New York in the following century. During the 19th century, most long-term U.S. bonds included

¹²During major political disruptions such as the revolutions of 1830 and 1848 and the Franco-Prussian War of 1870, credit spreads increased significantly.

an early redemption date, which effectively served as their maturity since the debt was generally repaid or refinanced as soon as allowed. As a result, the United States can be seen as having issued, over the first half of the century,¹³ a series of bonds with effective maturities ranging from 4 to 20 years.

An important specificity appears during the Civil War: bonds were issued with both an early redemption date and a final due date, often with a substantial gap between the two. This dual-dating structure clearly reflected the government's loss of confidence in its ability to refinance its debt on favorable terms at the earliest opportunity.

Following the Civil War and until the end of the 19th century, U.S. bond issuance remained diverse, featuring early redemption dates corresponding to a wide range of effective maturities and occasional issues of perpetual bonds. It is however difficult to draw a genuine U.S. yield curve in the 19th century.

1.2.6 From the Start of the 20th Century to the Present Day

The 20th century was marked by major geopolitical events – including two world wars – and significant economic upheavals such as the Great Depression of 1929, the high inflation that followed the postwar boom, and the rise of a second wave of globalization that continues into the 21st century. It was also a century of profound technological transformation. The pace of historical change accelerated to such a degree that any attempt to summarize the broader historical context in just a few paragraphs would be inevitably reductive.

However, in terms of debt markets, it is possible to identify the main structural trends that reshaped the nature of market participants and explain the major changes that, since the 1950s and 1960s, have made it possible to construct complete yield curves for the world's major economies and currencies.

One of the most significant developments in the 20th century, from the perspective of this paper, is the gradual replacement of the individual investor – embodied in the 19th-century and early 20th-century figures of the *rentier* and private speculator – by institutional asset managers. This shift, along with increased regulatory oversight, brought growing attention to asset-liability management and prudential concerns within banks, insurance companies, and pension funds. As a result, bond maturities diversified globally to meet the increasingly complex needs of institutional investors.

On the supply side, sovereign debt issuance surged due to the financing needs associated with the two world wars and, subsequently, to persistent deficits driven in part by the expansion of welfare states. Both France and the United Kingdom relied on a wide array of debt instruments – not only in terms of maturity but also with coupons indexed to inflation, gold, or foreign currencies – to maintain access to capital markets during challenging periods. In recent decades, regular issuance across a broad range of maturities has become the norm, enabling the construction of sovereign yield curves.

The share of perpetual bonds in European sovereign debt steadily declined throughout the 20th century, and investor demand eventually vanishes in the face of rising inflation. France still issued *rentes* for decades, but they disappeared from the financial landscape in 1982. British consols were finally redeemed in 2015. Only a few minor Dutch perpetuals remain outstanding. The old world inherited from medieval times has now disappeared.

In the United States, the creation of the Federal Reserve System helped stabilize short-term interest rates, which had been extremely volatile in the 19th century. From the 1920s onward, the U.S. government issued increasing volumes of short-term debt – initially in the form of Treasury certificates, later as 3-month Treasury bills. From the 1960s, regular issuance of Treasury bills with maturities of 3, 6, 9, and 12 months became standard.¹⁴

¹³In the 1830s, the long-term federal debt was largely paid down, though the government continued issuing short-term notes. Long-term issuance resumed in the 1840s.

¹⁴See <https://www.treasurydirect.gov/research-center/timeline/bills/> for recent data.

Beyond the short-term side of the curve, the United States issued a wide array of longer-dated instruments throughout the century. Initially, many of these included both an early redemption date and a final maturity date, often only a few years apart (especially from the 1930s onward). Over time, however, the U.S. transitioned to issuing fixed-maturity bonds across the full maturity spectrum. For more than 50 years now, Treasury bond issuance has been regular and highly structured,¹⁵ which, as in Europe, explains why U.S. yield curves have become a standard reference in global finance.

1.3 The Emergence of Cryptocurrencies

From ancient coinage bearing the faces of kings and emperors to modern fiat currencies managed by central banks, money has historically been intimately tied to a central authority if not the state itself. While money fulfills core economic functions – serving as a medium of exchange, a unit of account, and a store of value – it has also frequently operated as an instrument of governance, a symbol of regional or national sovereignty, and, at times, a tool of diplomacy or coercion.

The recent emergence of cryptocurrencies represents a significant departure from this historical pattern. Arising from the convergence of cryptography, computer science, and a form of libertarian thought, most cryptocurrencies propose a model of money that is neither issued by a central authority, nor backed by a government, nor governed by traditional monetary policy.¹⁶ Instead, cryptocurrencies are generally governed by open-source protocols with predefined rules, decentralized consensus mechanisms, and some form of cryptographic trust. This shift opens up the possibility – long theorized but never fully realized – of a monetary system independent of state control.

The intellectual lineage of this idea can be traced to Friedrich Hayek’s *The Denationalisation of Money* (1976), in which the famous Austrian economist advocated for a competitive market of privately issued currencies, independently from central banks and states. Although Hayek’s proposal was largely dismissed at the time as utopian or impractical, the advent of blockchain-based cryptocurrencies can be seen as a technological embodiment of his vision. Bitcoin, in particular, was introduced in the wake of the 2008 financial crisis as an alternative to state-backed fiat currencies and as a response to the perceived structural failures of the traditional banking system.

In the years that followed, Bitcoin inspired a growing ecosystem of other blockchains and cryptocurrencies with new technical and economic features. The launch of Ethereum in 2015 marked a major turning point. Ethereum goes indeed beyond peer-to-peer payments by introducing smart contracts – self-executing code snippets deployed on the blockchain – which enabled a new generation of decentralized applications (dApps). This development laid the foundations for what is now known as decentralized finance (DeFi), a set of protocols that aim to offer financial services without intermediaries (lending, borrowing, swapping, etc.).

Unlike in traditional finance, where the structure and depth of sovereign debt markets allow for the computation of yield curves for major currencies, it is not straightforward to associate a canonical term structure of interest rates with a cryptocurrency. Due to the pseudonymous nature of blockchain transactions, unsecured on-chain debt is in fact essentially unfeasible: credit risk must be mitigated not through legal enforcement, but through technical and economic safeguards embedded in smart contracts. As a result, the current existing forms of on-chain lending and borrowing resemble overcollateralized repurchase agreements (repos), in which the borrower must lock collateral of greater value than the amount borrowed.

¹⁵See <https://www.treasurydirect.gov/research-center/timeline/notes/> and <https://www.treasurydirect.gov/research-center/timeline/bonds/>.

¹⁶The link between cryptocurrency markets and traditional finance has been facilitated by the introduction of stablecoins. These digital assets are designed to maintain a stable value relative to a fiat currency, most commonly the US dollar. Various types of stablecoins exist, including centralized ones such as USDC and USDT, and decentralized alternatives such as DAI and USDS. Together, they represent a major segment of the cryptocurrency market, with a combined capitalization exceeding \$200 billion as of early 2025. Source: <https://www.coingecko.com> and <https://www.defillama.com>.

1.4 Interest Rates in the Cryptocurrency Landscape

Despite the absence of a standard yield curve, it is nevertheless possible to identify meaningful forms of risk-free-like interest rates within the cryptocurrency ecosystem.

1.4.1 Existing Proxies for Short-Term Risk-Free Rates

Unlike Bitcoin’s blockchain, which relies on proof-of-work (PoW), many alternative blockchains have adopted proof-of-stake (PoS) mechanisms.¹⁷ In PoS systems, network security and consensus are maintained by validators, who are selected to propose and confirm blocks in proportion to the amount of native tokens they have staked. Validators are compensated for the service they provide – namely, securing the network and validating transactions – through protocol-defined rewards. These rewards, typically paid in the same native token, can be interpreted as a form of interest: they represent the economic counterpart to the capital locked through staking. In this sense, PoS introduces an endogenous yield mechanism that functions analogously to a short-term risk-free rate – one determined not by monetary authorities, but by the internal logic of the protocol.¹⁸

Beyond staking – which applies only to the native tokens of PoS blockchains – another important source of interest rates in the cryptocurrency ecosystem lies in decentralized lending protocols such as Aave, Compound and Morpho. These platforms allow users to deposit tokens into liquidity pools and earn interest. These interest payments are funded by borrowers who draw from the same pools, typically against overcollateralized positions.¹⁹

Interest rates in these decentralized money markets are generally determined algorithmically, based on real-time supply and demand dynamics, collateral quality, and protocol-specific parameters (see [FH22] for Aave’s interest rate model, [LH19] for Compound’s, and [CSBS23, BBGD⁺25, BCT25] for alternative modeling approaches). As of early 2025, such rates are computed block by block, and lending typically occurs at variable rates with no fixed maturity. While not strictly risk-free, these rates provide a useful and practical proxy for short-term interest rates within the cryptocurrency ecosystem. Thanks to overcollateralization and robust liquidation mechanisms, the risk of default remains indeed limited – although bad debt can still occur in the event of sudden price movements.²⁰

The dynamics of these rates have been studied in various works (e.g. [CKV23]) and often appear disconnected from the short-term interest rates observed in traditional finance, as noted for example in [BDM⁺24].²¹

1.4.2 Going Beyond Short Term with Derivatives

Whether through staking or lending protocols,²² only short-term or very short-term interest rates can be directly observed in the current cryptocurrency landscape. Yet for any form of intertemporal valuation, a broader term structure is essential. The question thus remains open:

How can one discount future cash flows denominated in a given cryptocurrency?

The aim of this paper is to reverse the perspective and use existing prices of derivative instruments to extract implicit interest rates. While fixed-maturity debt markets remain largely absent from the cryptocurrency space, derivatives markets have developed rapidly. Instruments such as futures and call and put options are now actively traded at multiple maturities, offering a viable basis from which to infer interest rates for various cryptocurrencies and horizons.

¹⁷For Ethereum, the transition from PoW to PoS was finalized in 2022 with “The Merge.”

¹⁸Staking can be performed directly by individual token holders or indirectly through services such as Lido, which pool user funds to participate collectively in the validation process. See [CG24] and [GVKT24] for a detailed discussion of staking, including the growing importance of liquid staking in DeFi.

¹⁹See [GWPK20] for more colors on lending protocols.

²⁰As in all Web3 protocols, technological and cybersecurity risks further distance these rates from a truly risk-free benchmark.

²¹Recent studies suggest that monetary policy may still explain part of the variation in DeFi interest rates; see [BBN23].

²²Other sources of yield exist, but staking and lending protocols are the main sources of risk-free rate proxies.

The use of derivatives to infer interest rates is not new in the financial literature. The very concept of convenience yield is grounded in this idea, although typically applied to futures contracts (see, for example, [DT21] for a general discussion on convenience yields). The convenience yield of Bitcoin, for instance, has been estimated in several recent studies, including [BHM23], [HN22], and [SST23].

The use of options to infer interest rates echoes an old question raised by Brenner and Galai in [BG86], who investigated whether the pricing of newly listed American call and put options on the CBOE – which began operations in 1973 – was consistent with prevailing interest rates. Using data from the late 1970s, they proposed a method to extract implied interest rates from option prices via call-put parity – an approach that resonates with our own. Related ideas have since been explored from different perspectives in [Nar09] and [KM23]. See also [FSKV23] for a discussion on call-put parity for cryptocurrencies.

Section 2 provides an overview of the cryptocurrency ecosystem, with a particular emphasis on derivatives markets. We explain our focus on the Deribit platform and present descriptive statistics related to the trading of call and put options. In Section 3, we show how the traditional call-put parity can be used in an unconventional way to construct synthetic zero-coupon bonds and extract implied interest rates. We also describe in detail the minimization problem solved to estimate these rates. Section 4 presents the empirical results obtained.

2 Market Fragmentation and Platform Focus

The cryptocurrency ecosystem refers to the interconnected network of technologies, platforms, and participants that facilitate the creation, trading, and utilization of digital assets. It encompasses blockchain networks, centralized exchanges, DeFi applications, and a wide range of financial products designed for both retail and institutional investors.

2.1 A Fragmented Market

2.1.1 (De)Centralization and Regulatory Framework

Understanding the different market segments is crucial for grasping the complex structure of the cryptocurrency ecosystem. These segments can be categorized according to the degree of centralization and the level of regulatory oversight.

Traditional financial exchanges, such as the Chicago Mercantile Exchange (CME), the European Exchange (Eurex), and the Chicago Board Options Exchange with its newly-created digital branch (Cboe Digital), have recently opened to cryptocurrency trading. These exchanges are subject to stringent rules enforced by local financial authorities such as the Securities and Exchange Commission (SEC) in the United States, the Financial Conduct Authority (FCA) in the United Kingdom, and the Autorité des Marchés Financiers (AMF) in France. This framework ensures compliance with financial laws, protects users from fraud, and promotes transparency. Most of these exchanges focus exclusively on cash-settled derivative products that do not involve the delivery of cryptocurrencies and require non-crypto high-quality collateral.

Crypto-native centralized exchanges, such as Binance, Deribit, and Coinbase, emerged directly from within the cryptocurrency ecosystem and initially operated with minimal oversight from traditional financial authorities. While they are now required to comply with regulatory frameworks in most jurisdictions – such as MiCA in Europe – they remain significantly less regulated than traditional financial exchanges. These platforms offer turnkey solutions and a seamless user experience, including wallet creation, access to a broad range of cryptocurrency pairs, and a variety of leveraged financial products. However, they typically process trades off-chain, relying on their own internal infrastructure rather than directly on blockchain technology.

Fully decentralized applications – such as Uniswap, Aave, MakerDAO, Compound, Morpho, dYdX, etc. – operate directly on blockchain networks like Ethereum, Solana, or Polygon through smart contracts and use oracles to link blockchains to off-chain systems. Known as “on-chain protocols,” these platforms facilitate

spot and derivatives trading, as well as borrowing and lending, without relying on centralized intermediaries, thereby limiting counterparty risk.

2.1.2 The Specific Case of Derivatives

The cryptocurrency derivatives market offers a diverse range of products, structured with varying quotation, margining, and settlement mechanisms to accommodate different investor profiles. Crypto-native exchanges such as Deribit, Binance, and OKX, along with fully decentralized applications such as dYdX, Lyra, GMX, and Synthetix, among others, typically provide access to linear and/or inverse derivative contracts. Linear contracts are margined and settled in a reference stablecoin, whereas inverse contracts are margined and settled in the underlying cryptocurrency.²³ Traditional financial exchanges, such as CME, offer futures on cryptocurrency indices as well as options on these futures, margined and settled in USD.

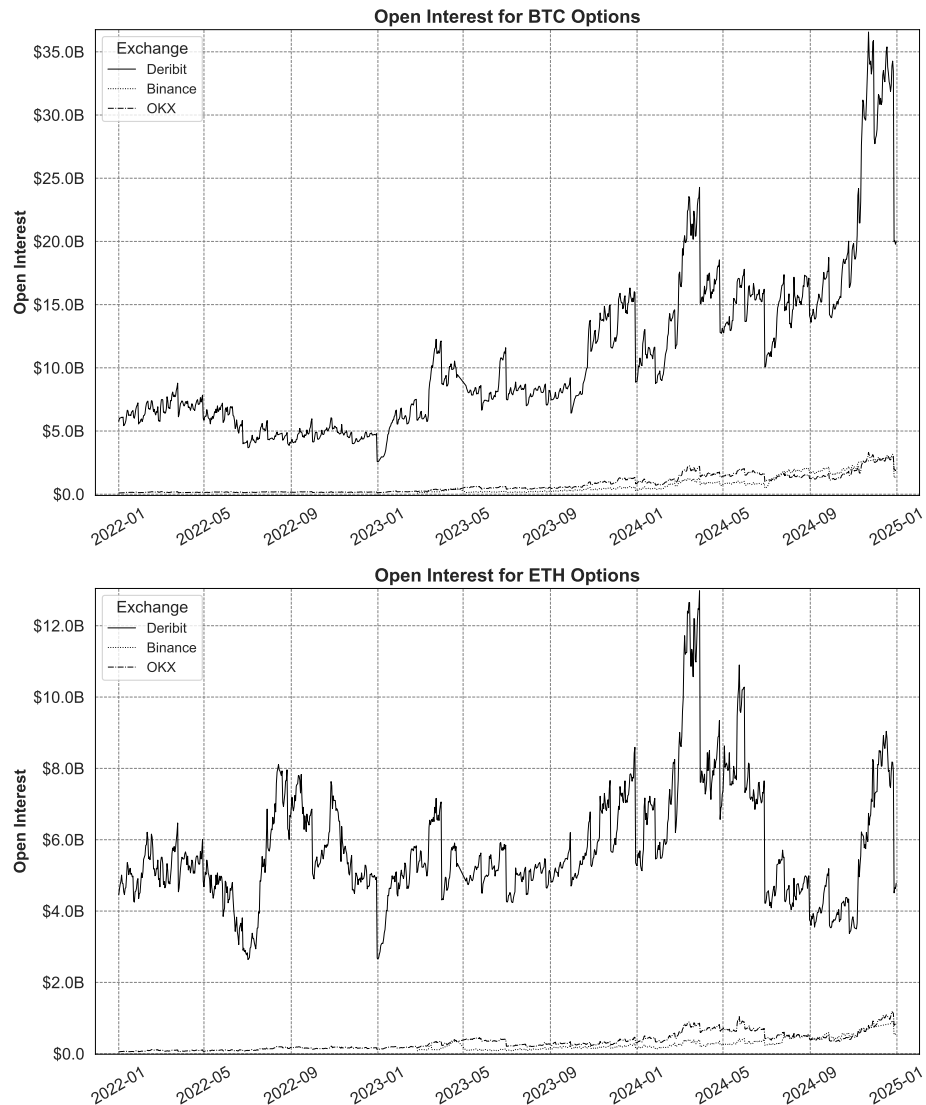


Figure 1: Daily aggregated open interest for options on BTC (top), and ETH (bottom) from January 1st, 2022 to December 31st, 2024. Source: <https://www.coinglass.com>.

²³For an interesting introduction to options on cryptocurrencies, see [CDA21].

Among crypto-native centralized platforms offering derivatives trading, Deribit has consistently dominated the market for options, maintaining over 80% of total open interest since 2022, as illustrated in Figure 1. This figure includes open interest for both BTC and ETH inverse options, the two largest cryptocurrencies by market capitalization.²⁴ In the decentralized space, the cumulative Total Value Locked (TVL) across all protocols offering derivatives trading is approximately \$100 million as of early 2025.²⁵ Given these figures and to avoid complications arising from fragmentation and the absence of standardized conventions across platforms, this paper focuses exclusively on Deribit data.

2.2 Derivatives on Deribit

Deribit offers fixed-maturity inverse futures and European inverse options on BTC and ETH. On Deribit, inverse options are quoted in the underlying currency, with strikes in the reference currency; inverse futures, however, are quoted directly in the reference currency.²⁶ While Deribit also lists linear perpetual futures, we do not consider them in this paper, which focuses exclusively on fixed-maturity products. In addition, linear options – quoted, margined, and settled in USDC – are available for BNB, PAXG, SOL and XRP, but their liquidity as of early 2025 is insufficient for our purposes.

A wide range of maturities is available, including day+1, day+2, and day+3 options, in addition to weekly, monthly, and quarterly expiries. Each derivative contract expires at 08:00 UTC on its maturity date. Weekly contracts expire on Fridays, while monthly and quarterly contracts expire on the last Friday of the corresponding month or quarter. Table 1 presents a snapshot (as of March 1st, 2024) of available maturities and the associated moneyness ranges for BTC inverse options listed on Deribit. Notably, for monthly and quarterly contracts, the exchange offered an especially wide spectrum of strikes. This structure facilitates long-term directional positioning and a variety of volatility-based strategies.

Expiry date	Days to expiry	Strike (USD)		Moneyness range	
		Min	Max	Min	Max
2024-03-02	1	52,000	72,000	85%	118%
2024-03-03	2	54,000	74,000	86%	121%
2024-03-04	3	54,000	75,000	86%	121%
2024-03-08	7	42,000	76,000	67%	125%
2024-03-15	14	42,000	76,000	67%	124%
2024-03-22	21	48,000	80,000	76%	131%
2024-03-29	28	23,000	120,000	37%	196%
2024-04-26	56	15,000	95,000	24%	155%
2024-05-31	91	20,000	100,000	32%	163%
2024-06-28	119	10,000	120,000	16%	197%
2024-09-27	210	10,000	160,000	16%	262%
2024-12-27	301	10,000	180,000	16%	295%

Table 1: Maturities and moneyness ranges for BTC inverse options listed on Deribit as of March 1st, 2024. Source: <https://coinmetrics.io>.

Figure 2 illustrates the distribution of top-of-book sizes, i.e. sizes at the best bid and ask prices, for BTC inverse options listed on Deribit with maturities around 3, 6 and 12 months, during the first week of July 2024. Top-of-book sizes are grouped by moneyness ranges. For the first quarterly maturity, out-of-the-money (OTM) contracts exhibit larger top-of-book sizes, with greater dispersion and numerous outliers.

²⁴ As of early 2025, the market capitalizations of BTC and ETH are approximately \$1.90 trillion and \$400 billion, respectively.

²⁵ Source: <https://defillama.com>.

²⁶ Deribit chose USD as the reference currency for these derivatives. A stablecoin would have been a more crypto-native alternative, but would have introduced the risk of the stablecoin depegging.

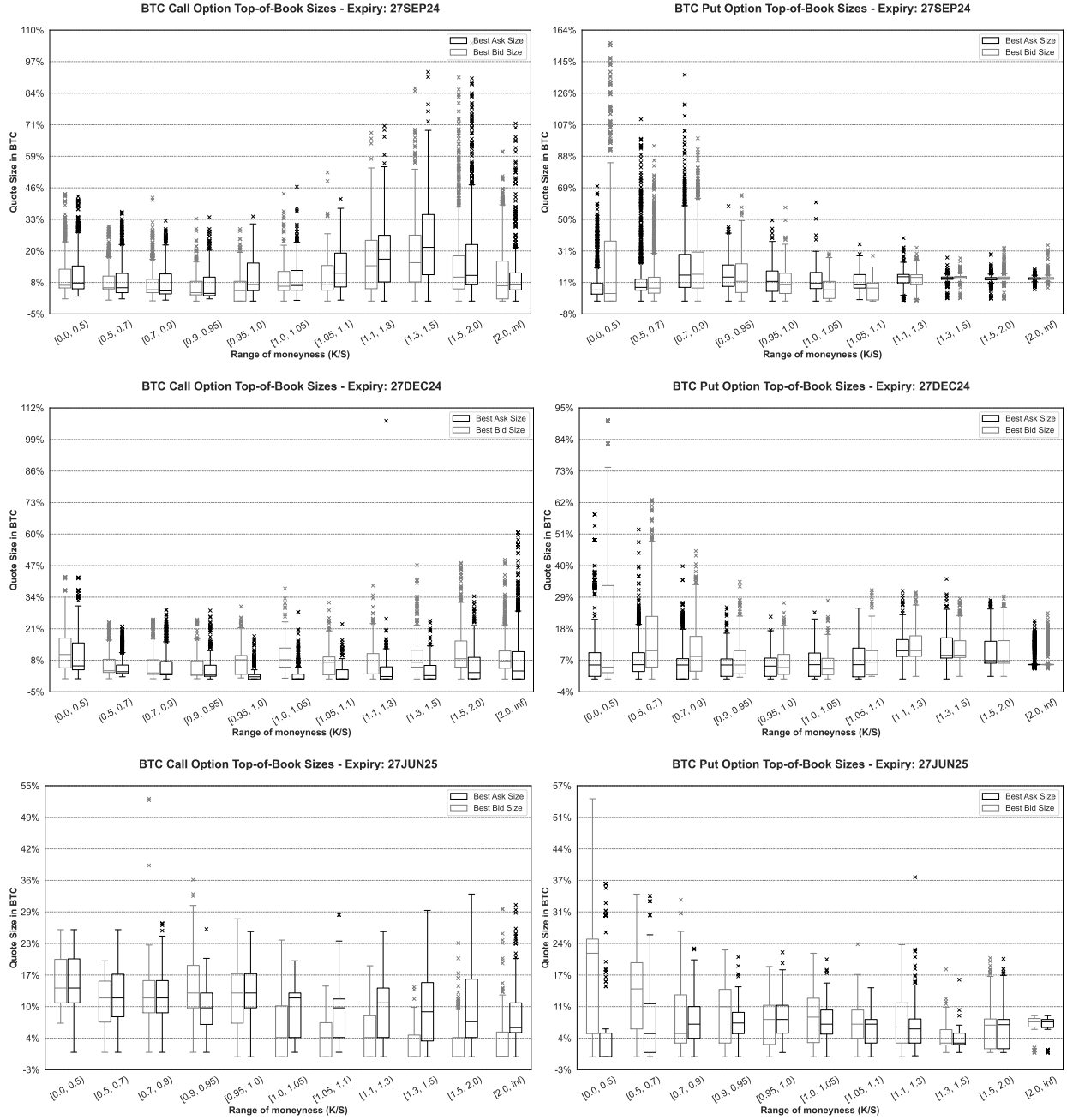


Figure 2: Boxplot of the top-of-book sizes of BTC inverse option contracts listed on Deribit for different maturities. Option expiries: September 27th, 2024 (top), December 27th, 2024 (middle), and June 27th, 2025 (bottom). Hourly data were collected from July 1st, 2024 to July 7th, 2024. Source: <https://coinmetrics.io>.

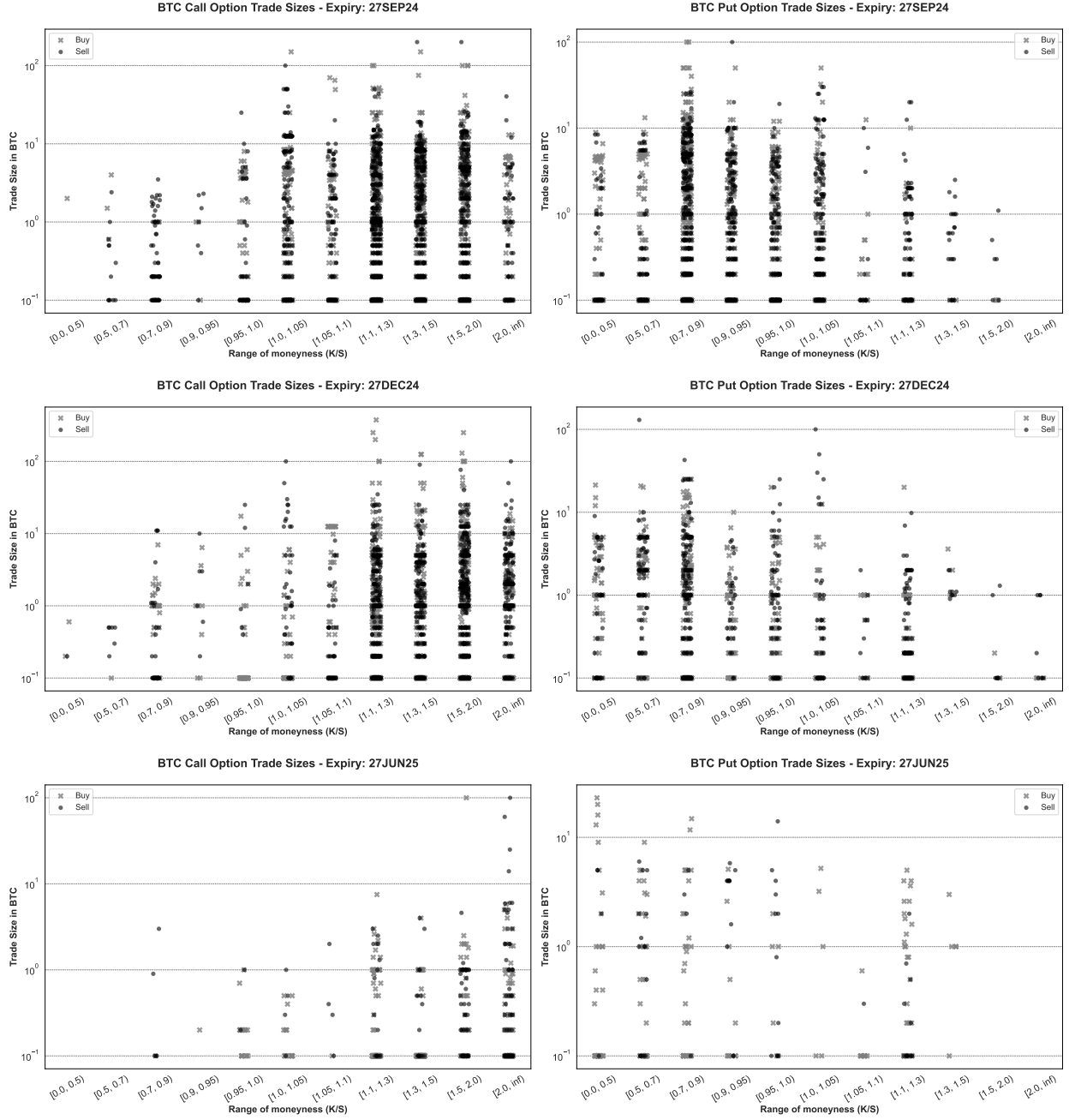


Figure 3: Distribution (logarithmic scale) of trade sizes of BTC inverse option contracts listed on Deribit for different maturities. Option expiries: September 27th, 2024 (top), December 27th, 2024 (middle), and June 27th, 2025 (bottom). Data from July 1st, 2024 to July 7th, 2024. Source: <https://coinmetrics.io>.

These top-of-book sizes are in line with trader's interests for the different contracts as confirmed by the transaction patterns shown in Figure 3, which displays the traded sizes of the same BTC inverse option contracts over the same week in July 2024.²⁷ OTM options are significantly more actively traded than in-the-money (ITM) options. This is expected because OTM options are frequently used by speculators betting on a directional move, notably trend followers, as they offer leverage at lower upfront cost. We also note the relatively lower trading activity in contracts with longer maturities.

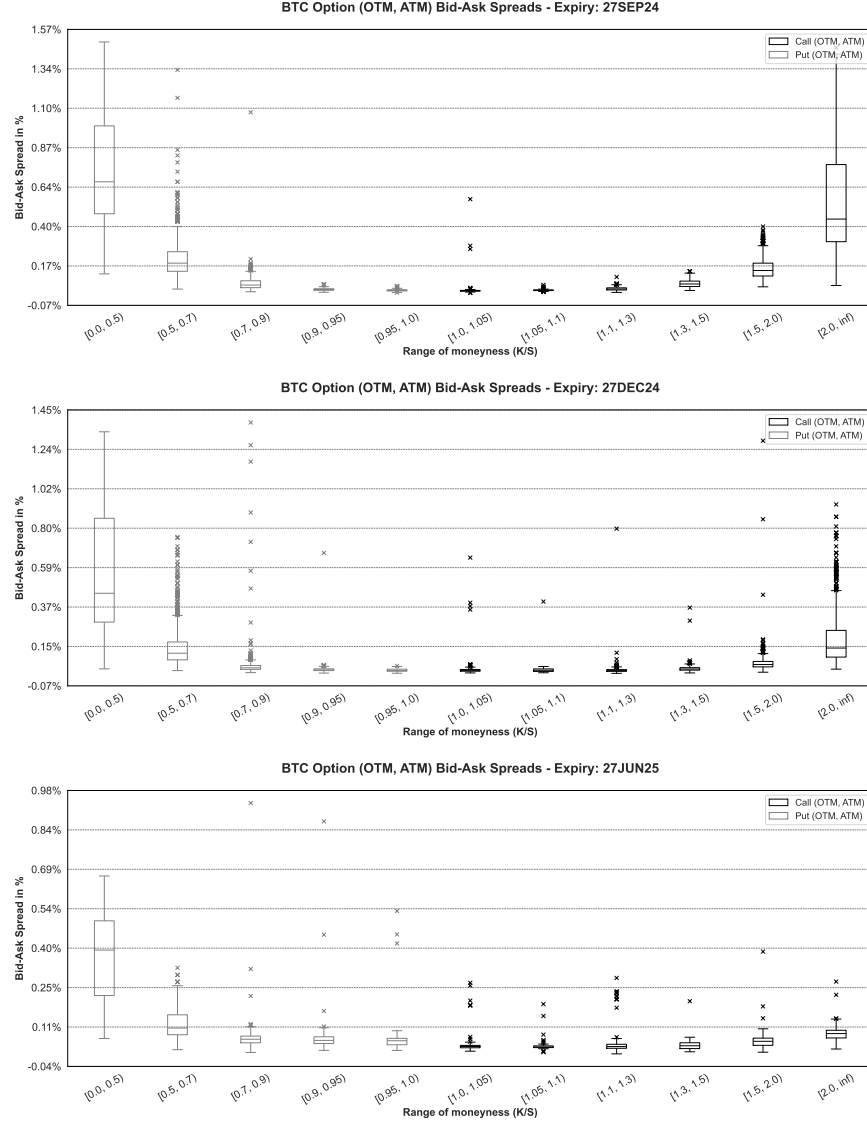


Figure 4: Boxplot of bid-ask spreads (in percentage) for BTC inverse option contracts listed on Deribit for different maturities. Option expiries: September 27th, 2024 (top), December 27th, 2024 (middle), and June 27th, 2025 (bottom). Only out-of-the-money and at-the-money options for both call and put contracts are included. Data from July 1st, 2024 to July 7th, 2024. Source: <https://coinmetrics.io>.

Figure 4 illustrates the bid-ask spread for ATM and OTM option contracts on BTC. The U-shaped pattern as a function of moneyness highlights higher liquidity near moneyness equal to 1, where bid-ask spreads are narrowest. In contrast, far OTM options exhibit wider spreads, reflecting lower liquidity and greater pricing uncertainty.

²⁷The minimum size on these options corresponds to 0.1 BTC.

3 Methodology

3.1 Back to First Principles

In the traditional framework of financial mathematics, the pricing of derivative products relies fundamentally on the assumption of absence of arbitrage opportunities. According to this assumption – linked to the “Law of One Price” – two financial instruments with identical payoffs must have the same price today. In particular, the theoretical price of a derivative contract is given by the initial cost of a self-financing trading strategy that exactly replicates its payoff.

For instance, the classical Black-Scholes formula can be derived from the dynamic replicating portfolio of a call or put option, which consists of the underlying asset and a risk-free asset. Similarly, in the case of a forward contract, the static replication strategy involves a combination of the underlying asset and the risk-free asset. In both cases, pricing formulas explicitly involve the price of a risk-free asset, or equivalently, a risk-free rate.

In the context of cryptocurrencies, the absence of fixed-income instruments from which to derive yield curves makes the situation seemingly more complex. However, the fundamental principles of mathematical finance still apply. Our approach is to reverse the classical reasoning and replicate the payoff of (non-existent) zero-coupon bonds using strategies involving existing cryptocurrency derivatives – specifically, inverse options and inverse futures with fixed maturities. From observed market prices of listed derivatives, one can therefore synthetically price zero-coupon bonds and thus construct yield curves for the cryptocurrencies on which these derivatives are written.

3.2 From the Replication of Zero-Coupon Bonds to Interest Rate Estimates

3.2.1 Revisiting the Call-Put Parity

In traditional finance, the call-put parity (which arises from basic static replication arguments) provides a classical relationship between the prices of a European (linear) call option, a European (linear) put option, a forward / (linear) futures contract,²⁸ and a zero-coupon bond.

Let us now turn to the case of a cryptocurrency (denoted by the symbol $\mathfrak{C}$) where the listed derivatives are inverse European call and put options, as well as inverse futures. We consider the following portfolio strategy:

- buying an inverse call option on $\mathfrak{C}$ with expiry date T and strike K (in the reference currency),
- selling an inverse put option on $\mathfrak{C}$ with expiry date T and strike K (in the reference currency),
- entering into a short position in an inverse futures contract on $\mathfrak{C}$ with expiry date T – we denote by $F_t(T)$ (in reference currency) the price of the contract.

By definition of the derivative contracts, if S_T is the price at date T of $\mathfrak{C}$ (in reference currency), then the portfolio at date T contains:

$$\underbrace{\left(1 - \frac{K}{S_T}\right)^+}_{\text{Payoff of the inverse call}} - \underbrace{\left(\frac{K}{S_T} - 1\right)^+}_{\text{Payoff of the inverse put}} - \underbrace{\left(1 - \frac{F_t(T)}{S_T}\right)}_{\text{Payoff of the inverse future}} = \frac{F_t(T) - K}{S_T} \quad \mathfrak{C}.$$

If $F_t(T) \geq K$, converting this amount of $\mathfrak{C}$ into the reference currency at date T (at price S_T) gives $F_t(T) - K \geq 0$ in the reference currency. In other words, this portfolio replicates the payoff of $F_t(T) - K$ zero-coupon bonds with maturity T in the reference currency.²⁹

²⁸Throughout this text, we ignore the influence of margin mechanisms.

²⁹It is as if these zero-coupon bonds were issued by the trading platform. Consequently, it is likely that zero-coupon bond prices vary across platforms, each probably incorporating its specific credit risk.

If $F_t(T) < K$, the trader owes the platform $K - F_t(T) > 0$ in the reference currency. In this case, the situation is equivalent to having replicated a short position in $K - F_t(T)$ zero-coupon bonds with maturity T .³⁰

As a result, the price $ZC_t^{\text{ref}}(T)$ (in reference currency) at time t of a zero-coupon bond in the reference currency maturing at T should satisfy:

$$C_t(K, T) - P_t(K, T) = \frac{F_t(T) - K}{S_t} ZC_t^{\text{ref}}(T) \quad (1)$$

where S_t is the price of € in the reference currency at time t , and $C_t(K, T)$ and $P_t(K, T)$ denote the respective prices (in €) of the inverse call and put options on € with strike K and maturity T .

3.2.2 Inverse Futures and Currency Conversion of Zero-Coupon Bonds

A similar line of reasoning as in the previous paragraphs can be applied to replicate a zero-coupon bond in € from a portfolio combining zero-coupon bonds in the reference currency (identified with the static replicating portfolio described above) and an inverse futures contract. More precisely, using the same notation as before, we consider the following portfolio strategy:

- buying $F_t(T)$ zero-coupon bonds in the reference currency maturing at T ,
- entering a long position in an inverse futures contract on € with expiry date T (futures price $F_t(T)$ in the reference currency).

At maturity T , assuming the proceeds of the zero-coupon bond are converted into € , the portfolio contains:

$$\underbrace{\frac{F_t(T)}{S_T}}_{\substack{\text{Payoff of the} \\ \text{zero-coupon bonds} \\ \text{converted in } \text{€}}} + \underbrace{\left(1 - \frac{F_t(T)}{S_T}\right)}_{\substack{\text{Payoff of the} \\ \text{inverse future}}} = 1 \quad \text{€}.$$

This portfolio therefore delivers one unit of € at time T , replicating the payoff of a zero-coupon bond in € with expiry date T . Consequently, the price $ZC_t^{\text{€}}(T)$ (in €) at time t of a zero-coupon bond in € maturing at T should be:

$$ZC_t^{\text{€}}(T) = \frac{F_t(T)}{S_t} ZC_t^{\text{ref}}(T), \quad (2)$$

where we divided by S_t because $ZC_t^{\text{€}}(T)$ is in € .

3.2.3 General Principles of Statistical Estimations

Eq. (1) corresponds to the classical call–put parity with forwards or futures, expressed in units of € . In practice, however, for a given time t and expiry date T , the ratio

$$\frac{C_t(K, T) - P_t(K, T)}{(F_t(T) - K)/S_t}$$

is not constant across strikes K . Naïve approaches – such as averaging this ratio over strikes or regressing the numerator on the denominator – yield poor statistical estimators. A more robust method is to jointly exploit Eq. (1) and Eq. (2) to stabilize the estimation.

More precisely, we estimate the two zero-coupon bond prices, $ZC_t^{\text{€}}(T)$ and $ZC_t^{\text{ref}}(T)$, by minimizing simultaneously:

1. the distance between $C_t(K, T) - P_t(K, T)$ and $ZC_t^{\text{€}}(T) - \frac{K}{S_t} ZC_t^{\text{ref}}(T)$ across all available strikes K , and

³⁰At the trade level, this implies that some capital is locked on the platform as margin – a global effect that we ignore, as in traditional finance, when pricing a single product.

2. the distance between $F_t(T) \text{ZC}_t^{\text{ref}}(T)$ and $S_t \text{ZC}_t^{\zeta}(T)$.

To keep the procedure simple and obtain estimators in closed form, we adopt as the objective function a weighted sum of squared Euclidean distances.

For each available cryptocurrency³¹ and each expiry date T , the estimated zero-coupon bond prices yield the corresponding interest rates for maturity $T - t$ at time t . We denote by $r_t^{T-t, \text{ref}}$ the rate for the reference currency, and by $r_t^{T-t, \zeta}$ the rate for ζ .

3.3 Estimation Procedure

For a fixed expiry date T and a given time $t < T$, a cryptocurrency options market typically offers multiple contracts with different strikes $(K^i)_{1 \leq i \leq n_t^T}$. The minimization problem described above can therefore be written as

$$\underset{\alpha_t, \beta_t \in \mathbb{R}^2}{\operatorname{argmin}} \underbrace{\sum_{i=1}^{n_t^T} (y_t^i - \alpha_t + \beta_t m_t^i)^2 + \lambda_{n_t^T} \left(\alpha_t \frac{F_t}{S_t} - \beta_t \right)^2}_{=: f(\alpha_t, \beta_t)}, \quad (3)$$

where the α_t and β_t variables represent the (unknown) zero-coupon bond prices written on the cryptocurrency and on the reference currency respectively, $y_t^i := C_t(T, K^i) - P_t(T, K^i)$ is the call-put price difference from Eq. (1) for the i^{th} strike, $m_t^i := \frac{K^i}{S_t}$ is the moneyness of the i^{th} strike, F_t is a shorthand for $F_t(T)$, and $\lambda_{n_t^T}$ is a weighting parameter that scales the relative importance of options and futures data.³²

When $\lambda_{n_t^T} = 0$, the estimation relies solely on option prices, which broadly determines the absolute level of the α_t and β_t variables. Including the second term, linked to futures prices, stabilizes the ratio $\frac{\alpha_t}{\beta_t}$ and, consequently, the difference between the interest rates associated with the underlying cryptocurrency and with the reference currency.

The above problem specification takes the form of the minimization of a polynomial of degree 2 in (α_t, β_t) . Subsequently, it yields closed-form expressions for the minimizers. Setting the gradient

$$\nabla f(\alpha_t, \beta_t) = \begin{bmatrix} 2\alpha_t \left(n_t^T + \lambda_{n_t^T} \frac{F_t^2}{S_t^2} \right) - 2\beta_t \left(\lambda_{n_t^T} \frac{F_t}{S_t} + \sum_{i=1}^{n_t^T} m_t^i \right) - 2 \sum_{i=1}^{n_t^T} y_t^i \\ -2\alpha_t \left(\lambda_{n_t^T} \frac{F_t}{S_t} + \sum_{i=1}^{n_t^T} m_t^i \right) + 2\beta_t \left(\lambda_{n_t^T} + \sum_{i=1}^{n_t^T} (m_t^i)^2 \right) + 2 \sum_{i=1}^{n_t^T} m_t^i y_t^i \end{bmatrix}$$

to zero leads indeed to a linear system of two equations in two unknowns, whose solution provides the minimizer (α_t^*, β_t^*) of f and therefore estimators of the prices of zero-coupon bonds:

$$\begin{aligned} \widehat{\text{ZC}}_t^{\zeta}(T) &:= \alpha_t^* = \frac{1}{d} \left[\left(\lambda_{n_t^T} + \sum_{i=1}^{n_t^T} (m_t^i)^2 \right) \sum_{i=1}^{n_t^T} y_t^i - \left(\lambda_{n_t^T} \frac{F_t}{S_t} + \sum_{i=1}^{n_t^T} m_t^i \right) \sum_{i=1}^{n_t^T} m_t^i y_t^i \right], \\ \widehat{\text{ZC}}_t^{\text{ref}}(T) &:= \beta_t^* = \frac{1}{d} \left[\left(\lambda_{n_t^T} \frac{F_t}{S_t} + \sum_{i=1}^{n_t^T} m_t^i \right) \sum_{i=1}^{n_t^T} y_t^i - \left(n_t^T + \lambda_{n_t^T} \frac{F_t^2}{S_t^2} \right) \sum_{i=1}^{n_t^T} m_t^i y_t^i \right], \end{aligned}$$

where

$$d = \left(\lambda_{n_t^T} + \sum_{i=1}^{n_t^T} (m_t^i)^2 \right) \left(n_t^T + \lambda_{n_t^T} \frac{F_t^2}{S_t^2} \right) - \left(\lambda_{n_t^T} \frac{F_t}{S_t} + \sum_{i=1}^{n_t^T} m_t^i \right)^2.$$

³¹Because some cryptocurrencies (or the reference currency) may be common to several derivatives, one could also design a more sophisticated estimation procedure leveraging data from multiple underlyings to jointly infer zero-coupon bond values.

³²Here we use the same weight for all available options, but far in- or out-of-the-money options may be more often excluded when cleaning the data (cf. *infra*).

The corresponding interest rate estimators are given by:

$$\hat{r}_t^{T-t,\zeta} := -\frac{1}{T-t} \log \left(\widehat{ZC}_t^\zeta(T) \right) \quad \text{and} \quad \hat{r}_t^{T-t,\text{ref}} := -\frac{1}{T-t} \log \left(\widehat{ZC}_t^{\text{ref}}(T) \right).$$

Although the formulas above are theoretically straightforward to compute, cryptocurrency option market data are typically very noisy, requiring additional steps (beyond standard data cleansing) to obtain reliable estimators.

First, regarding data frequency, we aggregate hourly observations into daily estimates of zero-coupon bond prices and interest rates. This choice rests on the reasonable assumption that cryptocurrency interest rates evolve more slowly than cryptocurrency prices themselves.

Second, we implement an outlier detection and data trimming procedure based on a classical method from robust statistics: Random Sample Consensus (RANSAC; see [FB81]). RANSAC is an iterative, stochastic algorithm that estimates model parameters by fitting an ordinary least squares (OLS) regression on randomly selected subsets of the data. After each fit, observations are classified as *inliers* – those within a predefined residual threshold – or *outliers* – those outside it. The algorithm then selects the model that both fits the data well and maximizes the number of inliers.

It is worth noting that we do not apply the RANSAC algorithm directly to our initial estimation problem. Instead, to remove outliers and nontrivial anomalies in option prices, we run RANSAC at each time t and for each expiry date $T > t$ on the following regression model:

$$P_t^{\text{ask}}(T, K^i) - C_t^{\text{bid}}(T, K^i) = \zeta_t + \xi_t (C_t^{\text{ask}}(T, K^i) - P_t^{\text{bid}}(T, K^i)) + \eta_t^i, \quad (4)$$

In Eq. (4), the absolute value of the slope, i.e. $|\xi_t|$, reflects the relative strength of buying versus selling pressures, while the intercept ζ_t serves as a proxy for market liquidity. The intercept should remain positive, as a negative value would signal an arbitrage opportunity. Together, the slope and intercept provide a dynamic snapshot of market conditions, which can vary significantly over time. In particular, once the RANSAC algorithm has permitted to remove outliers for a given pair (t, T) ,³³ we discard all observations for that pair (t, T) if the estimated slope $\hat{\xi}_t$ deviates significantly from -1 .

In summary, for each date t and maturity $T > t$, we first apply RANSAC to Eq. (4) to identify and remove outliers. If the slope estimated from the inlier points differs significantly from -1 , we drop the pair (t, T) from the analysis. Otherwise, we proceed with the minimization procedure described above to obtain the zero-coupon bond price and interest rate estimators.

4 Empirical Analysis

As previously discussed, Deribit dominates the cryptocurrency options market, which justifies the use of their market data in the following analysis to derive interest rates using the approach described above.

4.1 Market Data

The dataset consists of Deribit’s options top-of-book, futures top-of-book, and BTC and ETH index prices sourced from the data provider CoinMetrics. Data is recorded at one-hour intervals from January 1st, 2022, to December 31st, 2024. This period captures significant market events, such as the Terra-LUNA crash in May 2022, the FTX bankruptcy in November 2022, the SVB crash in March 2023, the publication of the European regulation (MiCA) in June 2023, the Bitcoin halving in April 2024, and the U.S. elections followed by a bull run in 2024 (see Figure 5). For each data point, a mid-price is calculated as the average of the best bid and best ask prices from the order book.

³³ As a stochastic method, RANSAC requires careful implementation and careful tuning of key parameters, particularly the residual threshold.

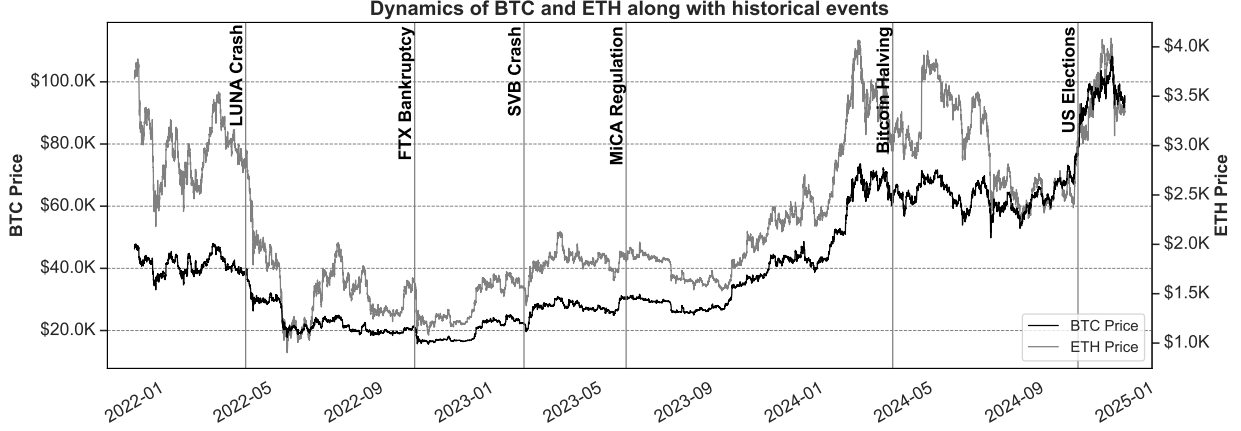


Figure 5: BTC and ETH prices from 1st January 2022 to 31st December 2024 along with historical events. Source: <https://coinmetrics.io>.

4.2 Data Trimming

Before applying our estimation procedure, the raw options data is filtered to ensure that the data points are representative of actual market dynamics.

First, options with maturities of less than one month are excluded. These short-term maturities tend to introduce excessive noise, often leading to unreasonable estimates, such as very negative interest rates. Second, we exclude data points with excessive bid-ask spreads with respect to the mid-price. Such data points do not accurately represent genuine buying or selling intentions, thus compromising data reliability. Then only, we use the trimming method based on the RANSAC algorithm described in Section 3.3.

The RANSAC method was applied to the full dataset of BTC inverse options, using a residual squared threshold of 0.004 to classify individual points and a tolerance threshold of 10% for ξ_t . Table 2 reports key descriptive statistics for the trimmed dataset, which are consistent with expectations: the average slope is very close to -1 , and the intercept is positive, small in magnitude, and associated with low standard deviations in both cases. These results are in line with theoretical predictions and indicate a reliable fit with limited variability. Similar figures are obtained for ETH inverse options.

	Mean	Std	Min	25%	50%	75%	Max
Slope	-1.01	0.02	-1.10	-1.01	-1.01	-1.00	-0.90
Intercept	0.02	0.02	0.00	0.01	0.02	0.03	0.51

Table 2: Descriptive statistics of the RANSAC estimates for the slope and intercept based on BTC inverse options from 1st January 2022 to 31st December 2024.

We illustrate the RANSAC method on May 30th, 2024, across quarterly, semi-annually, and annually expiry series of BTC inverse options. The top graphs in Figure 6 illustrates the RANSAC line distinguishing inliers from outliers. We also compute the algebraic distance between each data point and its orthogonal projection on the fitted RANSAC line. This distance captures both magnitude and direction (above or below the RANSAC line), as reflected in the distribution and cumulative plots.

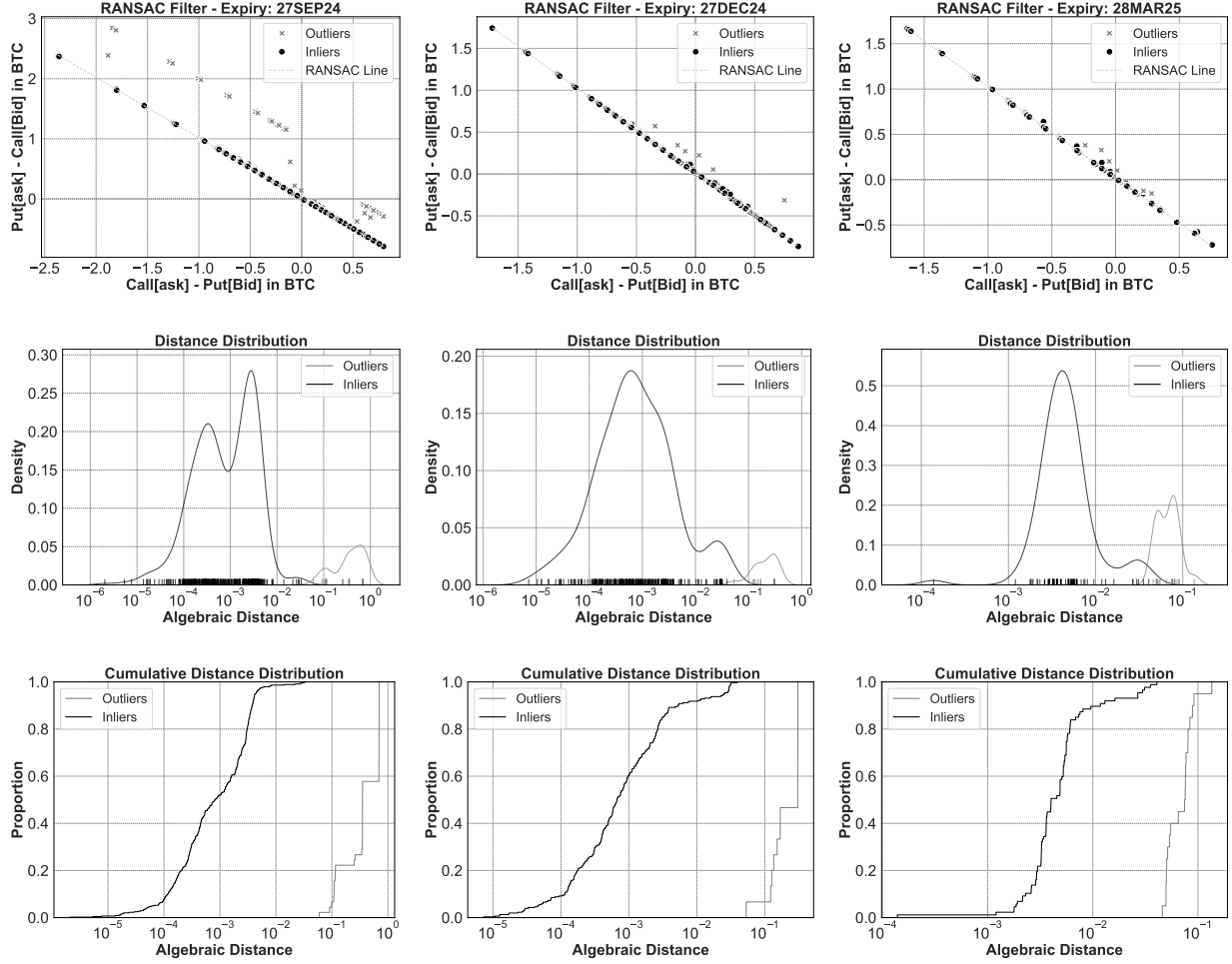


Figure 6: Detection of outliers using the RANSAC algorithm on 30th May, 2024. Expiry dates: September 27th, 2024 (left), December 27th, 2024 (middle), and March 28th, 2025 (right).

4.3 Results

We now present the interest rates and yield curves computed by applying our methodology to BTC and ETH inverse options traded on Deribit.

4.3.1 Implied Yield Curves

Using data on BTC inverse options, we compute, for a given date t , the interest rate estimates $\hat{r}_t^{T-t, \text{USD}}$ and $\hat{r}_t^{T-t, \text{BTC}}$ for each T in the set of available option maturities at date t . Figure 7 presents the resulting Deribit USD yield curve (left panel) and the BTC yield curve (right panel) on November 5th, 2024.

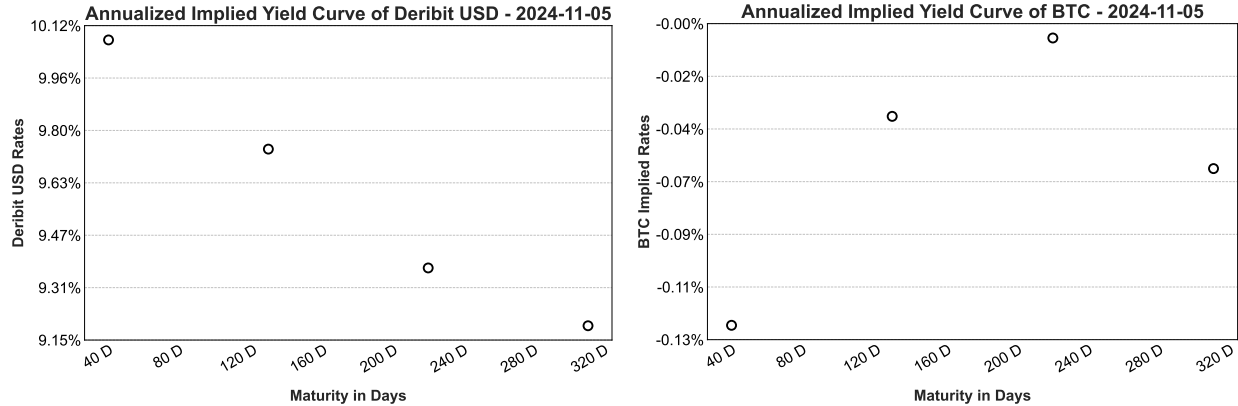


Figure 7: Annualized yield curves for Deribit USD (left) and BTC (right) computed using BTC inverse option data on November 5th, 2024.

Similarly, using data on ETH inverse options, we compute, for a given date t , the interest rate estimates $\hat{r}_t^{T-t, \text{USD}}$ and $\hat{r}_t^{T-t, \text{ETH}}$ for each T in the corresponding set of available maturities. Figure 8 presents the resulting Deribit USD yield curve (left panel) and the ETH yield curve (right panel) on November 5th, 2024.

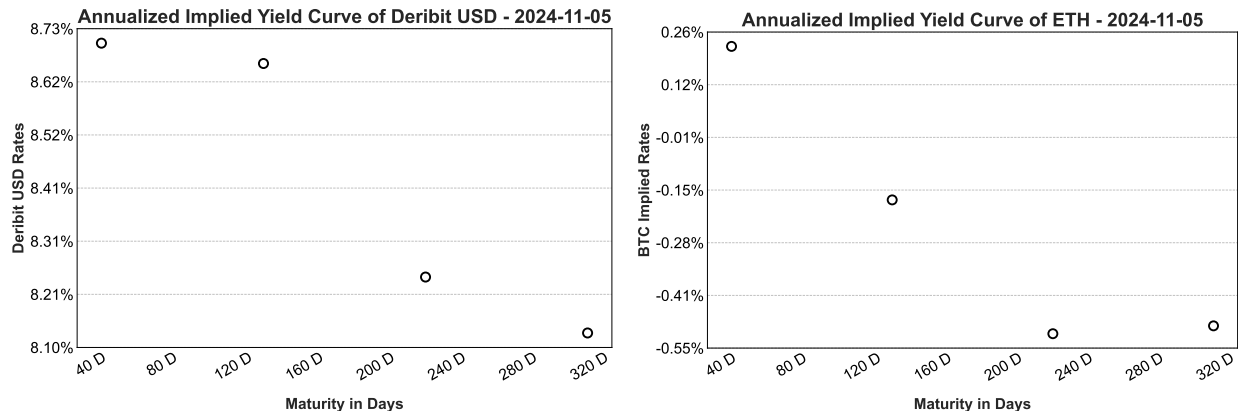


Figure 8: Annualized yield curves for Deribit USD (left) and ETH (right), computed using ETH inverse option data on November 5th, 2024.

The chosen date coincides with the announcement of Donald Trump's election at the end of 2024. The resulting downward-sloping structure suggests that traders may have been seeking exposure through near- and mid-term options. We expect the USD yield curves extracted from BTC and ETH inverse options to be broadly consistent with one another. Although this consistency is not immediately evident from the plots, the magnitudes of the inferred USD implied rates are roughly similar across the two markets – a

point examined in more detail in the next section. For BTC, the implied rates are close to zero, which is consistent with the fact that BTC does not intrinsically bear interest – although lending platforms, after conversion to wrapped tokens such as WBTC, can generate interest payments. In contrast, the low implied rates for ETH are somewhat surprising, given the existence of a non-negligible staking yield, not to mention additional returns available through lending platforms.

4.3.2 Implied Yield Curves Through Time

To provide a broader view of the yield curves obtained through our methodology, we construct their temporal evolution across different fixed maturities.

Figure 9 displays the time series of the Deribit USD and BTC yield curves for three tenors: 90 days, 180 days, and 360 days. Each point on these curves is computed by applying our methodology to BTC inverse option data, followed by linear interpolation from the nearest available maturities corresponding to the target tenor. Since Deribit often lists options with maturities slightly beyond 360 days, it is generally possible to recover the full set of displayed maturities (dates for which this was not possible were excluded).

The overall shapes of the curves reflect distinct market regimes. For example, in November 2022 – a highly volatile period marked by the collapses of FTX and Terra LUNA – implied rates for Deribit USD were very low, and in some cases, negative. Subsequently, Deribit USD rates rose steadily, reaching above 20% at certain points in 2024, before stabilizing between 10% and 15%. In the case of Deribit USD, changes in yield curve levels clearly dominate variations in slope and curvature.

The estimated BTC rates, shown in the right panel of Figure 9 (see also Table 3), remain close to zero over time across all tenors, apart from a brief dip into negative territory. These results indicate that the derivatives market is consistent with BTC carrying no intrinsic yield, notwithstanding the DeFi possibilities mentioned above.

	Mean	Std	Min	25%	50%	75%	Max
Tenor 90 days	0.0007	0.0143	−0.1011	−0.0016	0.0002	0.0031	0.1339
Tenor 180 days	0.0004	0.0056	−0.0485	−0.0012	0.0002	0.0022	0.0419
Tenor 360 days	−0.0001	0.0066	−0.0798	−0.0014	0.0000	0.0014	0.0608

Table 3: Distribution of BTC interest rates for the 3 different maturities (90 days, 180 days, and 360 days) computed with the Deribit BTC inverse option dataset from 1st January 2022 to 31st December 2024.

Figure 10 displays the time series of the Deribit USD and ETH yield curves for the same three tenors. Each point on these curves is obtained by applying our methodology to ETH inverse option data, followed by linear interpolation from the nearest available maturities corresponding to the target tenor. The implied rates for ETH fluctuate around zero throughout the observation period (see also Table 4). This result is puzzling, as one might expect interest rates to reflect, at least in part, the Ethereum staking yield, which has typically been around 2%.

	Mean	Std	Min	25%	50%	75%	Max
Tenor 90 days	−0.0023	0.0190	−0.2333	−0.0023	−0.0001	0.0014	0.1720
Tenor 180 days	−0.0015	0.0067	−0.0598	−0.0023	−0.0002	0.0010	0.0224
Tenor 360 days	−0.0014	0.0158	−0.3061	−0.0029	0.0000	0.0012	0.0671

Table 4: Distribution of ETH interest rates for the 3 different maturities (90 days, 180 days, and 360 days) computed with the Deribit ETH inverse option dataset from 1st January 2022 to 31st December 2024.

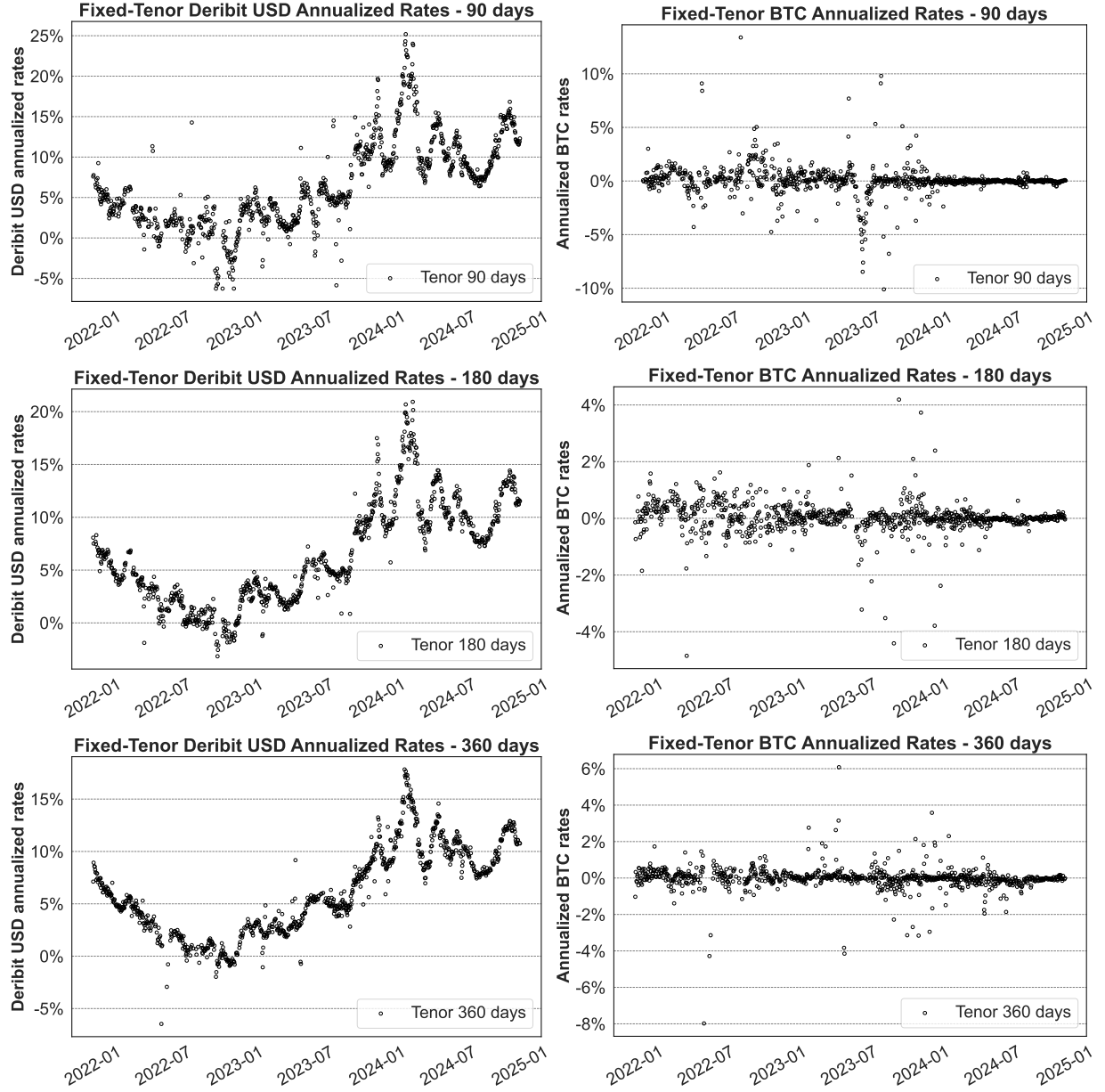


Figure 9: Time series of Deribit USD interest rates (left) and BTC interest rates (right). Maturities: 90 days (top), 180 days (middle), and 360 days (bottom) – Deribit BTC inverse option dataset from 1st January 2022 to 31st December 2024.

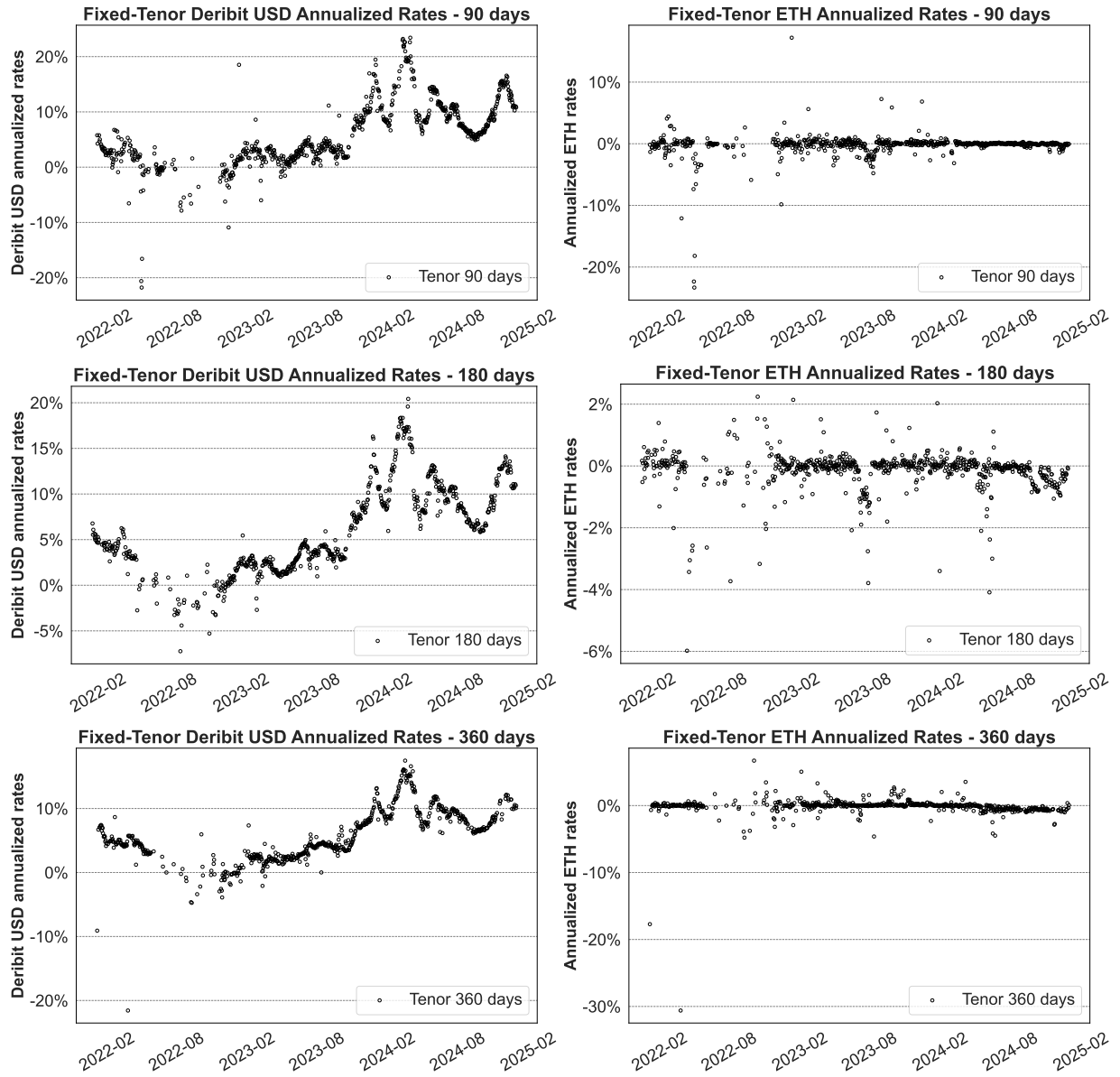


Figure 10: Time series of Deribit USD interest rates (left) and ETH interest rates (right). Maturities: 90 days (top), 180 days (middle), and 360 days (bottom) – Deribit ETH inverse option dataset from 1st January 2022 to 31st December 2024.

Although Deribit USD interest rates exhibit similar behavior and comparable magnitudes across the two cases, Table 5 shows that the differences between USD rates inferred from BTC and ETH inverse option data do not average to zero. Instead, they are frequently positive, with an average spread of approximately 1%, which tends to decrease with the tenor. Understanding the origin of this persistent discrepancy is an important direction for further investigation.

	Mean	Std	Min	25%	50%	75%	Max
Tenor 90 days	0.0105	0.0239	-0.1507	0.0008	0.0102	0.0176	0.2455
Tenor 180 days	0.0092	0.0099	-0.0212	0.0042	0.0086	0.0145	0.0711
Tenor 360 days	0.0081	0.0154	-0.0347	0.0024	0.0069	0.0127	0.2687

Table 5: Difference in USD Deribit rates inferred from BTC and ETH inverse options data on the Deribit dataset from 1st January 2022 to 31st December 2024.

4.3.3 Implied Yield Curves and Aave Rates

One of the puzzling outcomes of our results is that the implied interest rates for the two true cryptocurrencies – BTC and ETH – remain close to zero across maturities. While this finding is somewhat expected for BTC, which does not natively bear interest, it is more surprising in the case of ETH, which offers a non-negligible staking yield. In contrast, the implied Deribit USD rates inferred from both BTC and ETH options appear broadly consistent with expectations.

To assess more precisely the economic relevance of these USD implied rates within the cryptocurrency ecosystem, we compare them with the lending rates observed on the DeFi protocol Aave, which pays interest on deposited assets through a dynamic rate mechanism. Although the credit profile of a stablecoin differs from that of a USD-denominated claim on a centralized platform, we use annualized instantaneous USDC lending rates from Aave v3 as a benchmark, over the period from February 6th, 2023 to December 31st, 2024.

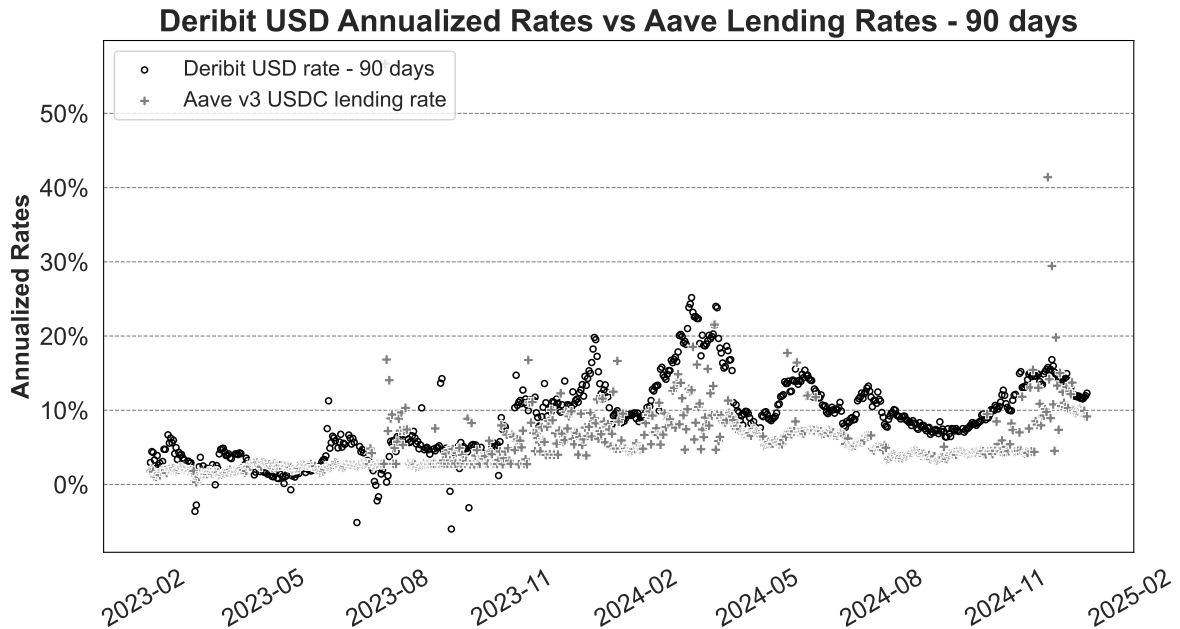


Figure 11: Comparison between Deribit USD rates obtained with our estimation procedure applied to BTC inverse option data and Aave v3 USDC lending rates, over the period from February 6th, 2023 to December 31st, 2024. Source of Aave v3 rates: <https://www.defillama.com>.

Figure 11 compares the temporal evolution of the (instantaneous) Aave USDC rates with the Deribit USD rate for a 90-day tenor, obtained through our estimation procedure applied to BTC inverse option data. We observe a general alignment between the two series, particularly during phases of pronounced upward or downward trends, although their absolute levels do not coincide. This comparison supports the economic relevance of the implied rates extracted through our methodology, while also underscoring that the different markets involved – centralized options and decentralized lending markets – are not yet fully integrated.

5 Conclusion

This paper presents a first step toward a systematic framework for estimating implied interest rates in the cryptocurrency ecosystem. In a bondless market that lacks a meaningful proxy for a risk-free rate – except in the very short term – we propose the construction of synthetic zero-coupon curved derived from derivative prices.

Our results indicate that the interest rates extracted from inverse options for a form of digital USD are broadly comparable to USDC lending rates on Aave, where the exposure to depegging risk is replaced by the bankruptcy risk of the lending platform. Importantly, these are not universal risk-free rates, but rather platform-specific rates. The estimates for BTC are broadly consistent with expectations, although likely still too low and too noisy. In contrast, the implied rates for ETH remain puzzling and require further investigation, particularly in light of Ethereum’s staking yield.

The framework can be refined and extended with other datasets. It also opens the door to cross-sectional analyses, such as comparing implied rates across trading venues. For example, the spread between a Deribit USD rate and a Binance USD rate could serve as a proxy for platform-specific risk. Comparisons with data from decentralized protocols such as dYdX, Lyra, GMX, or Synthetix could also provide valuable insights.

In this sense, the methodology developed here should be viewed as a foundation rather than a conclusion. Much remains to be explored, but it offers a concrete basis for addressing questions of pricing, risk, and valuation in a financial environment that operates without traditional fixed-income instruments.

Acknowledgment

We would like to express our sincere gratitude to all those who made possible the research that led to this paper. Special thanks go to Quentin Archer for his energy, tireless encouragement, and thoughtful comments. Warm thanks also go to Corentine Poilvet-Clédière for her trust in our project. We are also grateful to Philippe de Peretti for insightful discussions on econometric and statistical tools. The historical part further benefited from stimulating conversations with professionals from the industry.

Statement on Funding

The research conducted for this paper benefited from the financial support of the Research Initiative “Blockchain et Intelligence Artificielle pour les infrastructures de marché,” under the aegis of the Institut Europlace de Finance, in partnership with LCH SA. The authors bear sole responsibility for the content of this publication, which does not reflect the views or practices of LCH SA.

Statement on Potential Conflicts of Interest

Wenkai Zhang is an employee of LCH SA (the post-trade division of London Stock Exchange Group). Sébastien Bieber is a PhD student at Université Paris Dauphine–PSL, funded by the Research Initiative “Blockchain et Intelligence Artificielle pour les infrastructures de marché.”

Statement on Data Availability

The data that support the findings of this study are available from the authors upon reasonable request.

Statement on ChatGPT Usage

ChatGPT was used solely for language refinement and editing. All content, ideas, and analyses are original and remain the sole responsibility of the authors.

References

- [And01] Jean Andreau. La banque et les affaires dans le monde romain: Ive siècle av. j.-c.-iiiie siècle ap. j.-c. 2001.
- [Ash13] Thomas Southcliffe Ashton. *An economic history of England: The eighteenth century*. Routledge, 2013.
- [Bar87] Robert J. Barro. Government spending, interest rates, prices, and budget deficits in the United Kingdom, 1701–1918. *Journal of Monetary Economics*, 20(2):221–247, 1987.
- [BBGD⁺25] Charles Bertucci, Louis Bertucci, Mathis Gontier Delaunay, Olivier Gueant, and Matthieu Lesbre. Agents’ behavior and interest rate model optimization in defi lending. *Available at SSRN 4802776*, 2025.
- [BBN23] Andrea Barbon, Jean Barthélémy, and Benoît Nguyen. DeFi-ying the Fed? Monetary Policy Transmission to Stablecoin Rates. *SSRN Electronic Journal*, 2023.
- [BCT25] Bastien Baude, Damien Challet, and Ioane Muni Toke. Optimal risk-aware interest rates for decentralized lending protocols. *arXiv preprint arXiv:2502.19862*, 2025.
- [BD19] Dirk G. Baur and Thomas Dimpfl. Price discovery in bitcoin spot or futures? *Journal of Futures Markets*, 2019.
- [BDM⁺24] Jean Barthélémy, Clément Delaneau, Thomas Martignon, Benoît Nguyen, Marten Riho Pallum, and Salim Talout Zitan. Interest rates in decentralised finance. <https://www.banque-france.fr/en/publications-and-statistics/publications/interest-rates-decentralised-finance>, 2024.
- [BG86] Menachem Brenner and Dan Galai. Implied Interest Rates. *The Journal of Business*, 1986.
- [BHM19] Dorje C. Brody, Lane P. Hughston, and Bernhard K. Meister. Theory of Cryptocurrency Interest Rates. *SIAM Journal Financial Math*, 2019.
- [BHMx23] Arkorful Gideon Bruce, Chen Haiqiang, Gu Ming, and Liu Xiaoqun. What can we learn from the convenience yield of Bitcoin? Evidence from the COVID-19 crisis. *International Review of Economics & Finance*, 2023.
- [BM01] Damiano Brigo and Fabio Mercurio. *Interest rate models: theory and practice*, volume 2. Springer, 2001.
- [CDA21] Alexander Carol, Chen Ding, and Imeraj Arben. Crypto Quanto and Inverse Options. *Mathematical Finance*, 2021.
- [CG24] Sylvain Carré and Franck Gabriel. Liquid staking: When does it help? *Available at SSRN*, 2024.
- [CKV23] Amit Chaudhary, Roman Kozhan, and Ganesh Viswanath-Natraj. Interest Rate Parity in Decentralized Finance. *SSRN Electronic Journal*, 2023.

- [Coh53] Jacob Cohen. The element of lottery in british government bonds, 1694-1919. *Economica*, 20(79):237–246, 1953.
- [Coh97] Edward Cohen. *Athenian economy and society: A banking perspective*. Princeton University Press, 1997.
- [CSBS23] Samuel N Cohen, Leandro Sánchez-Betancourt, and Lukasz Szpruch. The economics of interest rate models in decentralised lending protocols. *Available at SSRN 4638390*, 2023.
- [DT21] William Diamond and Peter Van Tassel. Risk-Free Rates and Convenience Yields Around the World, 2021.
- [Dur42] David Durand. Basic yields of corporate bonds, 1900-1942. In *Basic Yields of Corporate Bonds, 1900-1942*, pages 1–40. NBER, 1942.
- [FB81] Martin A. Fischler and Robert C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981.
- [FH22] Emilio Frangella and Lasse Herskind. Aave v3 technical paper. 2022.
- [FSKV23] Nóra Felföldi-Szűcs, Balázs Králik, and Kata Váradi. Backward and Forward Put Call Parity in a Crypto Option Market – Evidence from Binance. *Finance Research Letters*, 2023.
- [Geh88] Adam K. Gehr. Undated futures markets. *Journal of Futures Markets*, 8(1):89–97, 1988.
- [Gra14] David Graeber. *Debt: The First 5,000 Years, Updated and Expanded*. Melville House, 2014.
- [GVKT24] Krzysztof Gogol, Yaron Velner, Benjamin Kraner, and Claudio Tessone. Sok: Liquid staking tokens (lst) and emerging trends in restaking. *arXiv preprint arXiv:2404.00644*, 2024.
- [GWPK20] Lewis Gudgeon, Sam M. Werner, Daniel Perez, and William J. Knottenbelt. DeFi Protocols for Loanable Funds: Interest Rates, Liquidity and Market Efficiency. (2006.13922), June 2020.
- [HdM02] Michael Hudson and Marc Van de Mierop. Debt and economic renewal in the ancient near east: A colloquium held at Columbia University, 2002.
- [HN22] Jimmy E. Hilliard and Julie T. D. Ngo. Bitcoin: Jumps, convenience yields, and option prices. *Quantitative Finance*, 22(11):2079–2091, 2022.
- [Hub92] Peter J. Huber. Robust Estimation of a Location Parameter. *Breakthroughs in Statistics*, pages 492–518, 1992.
- [JKL21] Zhengyang Jiang, Arvind Krishnamurthy, and Hanno N. Lustig. Beyond Incomplete Spanning: Convenience Yields and Exchange Rate Disconnect. *SSRN Electronic Journal*, 2021.
- [Kim22] S. Thomas Kim. Is it worth to hold bitcoin? *Finance Research Letters*, 44:102090, 2022.
- [KM23] Muoria Kamau and Ivivi J. Mwaniki. On a robust estimation of option-implied interest rates and dividend yields. *Cogent Economics & Finance*, 11(2):2260658, 2023.
- [KN15] Robert L. Kosowski and Salih N. Neftci. Cash Flow Engineering and Alternative Classes (Commodities and Hedge Funds). In Robert L. Kosowski and Salih N. Neftci, editors, *Principles of Financial Engineering (Third Edition)*, Academic Press Advanced Finance, pages 211–236. Academic Press, 2015.
- [Kno02] Michael S. Knoll. Put-Call Parity and the Law. *University of Pennsylvania Carey Law School Follow*, 2002.
- [LH19] Robert Leshner and Geoffrey Hayes. Compound: The money market protocol. *White Paper*, 89, 2019.

- [MT97] Nikolaos T. Milonas and Stavros B. Thomadakis. Convenience yields as call options: An empirical analysis. *Journal of Futures Markets*, 17(1):1–15, 1997.
- [Mun12] John Munro. Usury, calvinism and credit in protestant england: from the sixteenth century to the industrial revolution. *Religion and religious institutions in the European economy. 1000-1800: Atti della Quarantatreesima Settimana di Studi, 8-12 maggio 2011*, pages 155–184, 2012.
- [Nar09] Lorenzo Naranjo. Implied Interest Rates in a Market with Frictions. *SSRN Electronic Journal*, 2009.
- [OW15] Akihiro Omura and Jason West. Convenience yield and the theory of storage: Applying an option-based approach. *Australian Journal of Agricultural and Resource Economics*, 59(3):355–374, 2015.
- [SH13] Richard Sylla and Sidney Homer. *A history of interest rates*. Wiley, 2013.
- [SL24] Artur Sepp and Vladimir Lucic. Valuation and hedging of cryptocurrency inverse options. *SSRN Electronic Journal*, 2024.
- [SST23] Maik Schmeling, Andreas Schrimpf, and Karamfil Todorov. Crypto Carry. *BIS Working Papers*, no. 1087, 2023.
- [Sto22] Harry Stobart. Pricing Interest Rate Swaps in Traditional and Decentralised Finance. *SSRN Electronic Journal*, 2022.
- [Tem04] Peter Temin. Financial intermediation in the early roman empire. *The Journal of Economic History*, 64(3):705–733, 2004.
- [The92] Henri Theil. A Rank-Invariant Method of Linear and Polynomial Regression Analysis. In Baldev Raj and Johan Koerts, editors, *Henri Theil's Contributions to Economics and Econometrics: Econometric Theory and Methodology*, pages 345–381. Springer Netherlands, Dordrecht, 1992.
- [Wes12] Jason West. Convenience Yields in Bulk Commodities: The Case of Thermal Coal. *SSRN Electronic Journal*, 2012.
- [WXZC21] Jinghong Wu, Ke Xu, Xinwei Zheng, and Jian Chen. Fractional cointegration in bitcoin spot and futures markets. *Journal of Futures Markets*, 41(9):1478–1494, 2021.